

Name: _____

GCSE Biology Unit 2 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries

Year	E	D	C	B	A	A*	Max
2018	7	10	16	25	34	45	56
2019	14	18	24	32	40	47	54
2022	6	8	16	26	36	47	58
2023	9	13	20	30	40	50	60

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3400UB0-1



BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms
HIGHER TIER

TUESDAY, 15 MAY 2018 – AFTERNOON
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	13	
3.	6	
4.	11	
5.	4	
6.	5	
7.	5	
8.	9	
9.	9	
10.	11	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **3** is a quality of extended response (QER) question where your writing skills will be assessed.



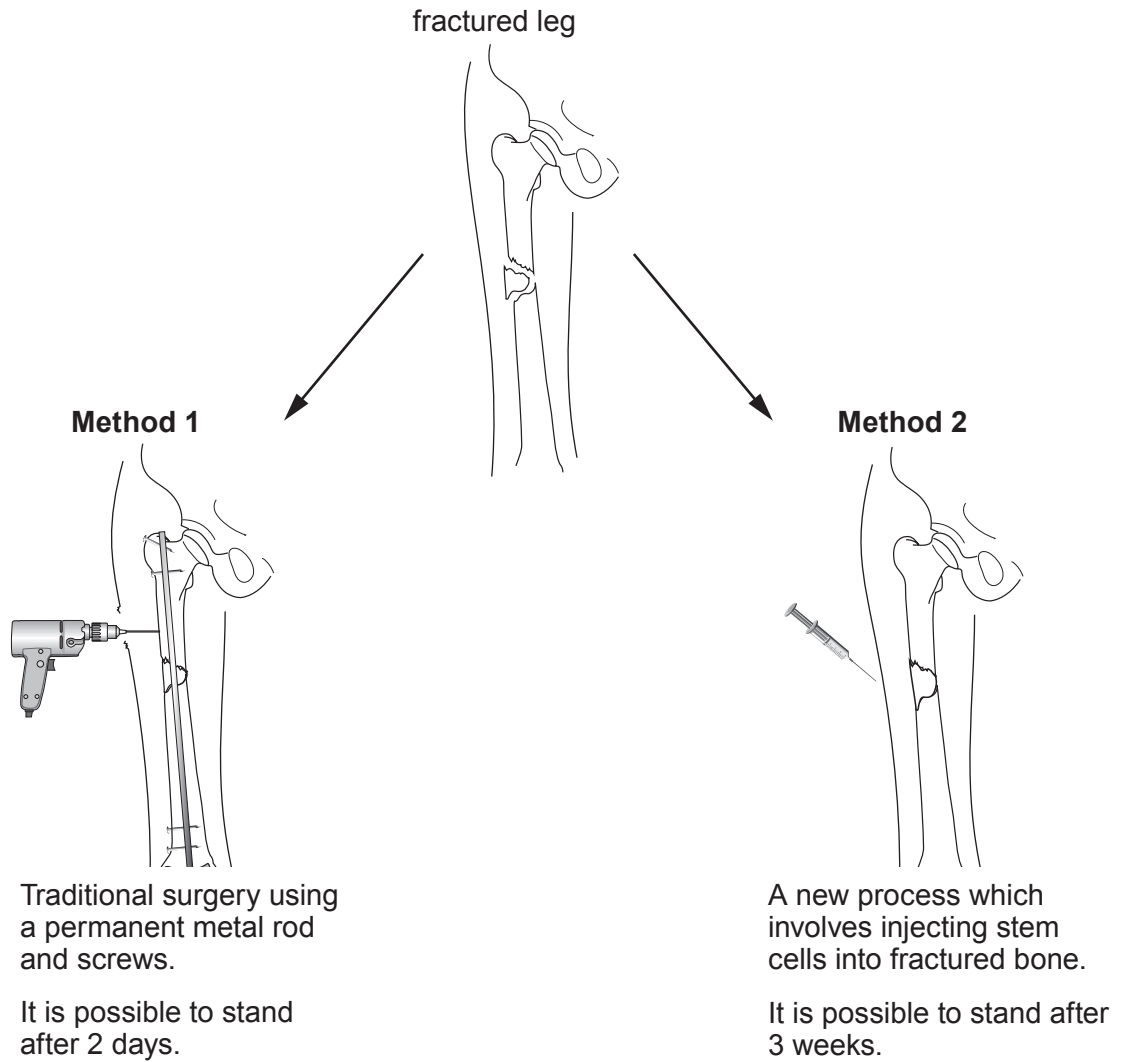
MAY183400UB0101

Answer all questions

1. The photograph below shows a young ice skater, Tracey, who falls and fractures her leg.



The diagram below shows two ways in which Tracey’s fractured leg could be treated in a large modern hospital.



Expected results of treatment

Time after treatment (weeks)	Percentage of bone healing (%)	
	Traditional surgery	Injection of stem cells
10	10	12
20	14	25
30	19	38
40	28	55
50	41	70
60	59	82

- (a) Give **one** reason why Tracey might choose traditional surgery and **one** reason why, alternatively, she might prefer to be treated by an injection of stem cells. [2]

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- (b) (i) When fractured bones heal, cells divide and multiply by mitosis.

Complete the table below.

[2]

Mitosis in human cells	
chromosomes in mother cell	46
number of daughter cells produced after one division	
number of chromosomes in daughter cells	

- (ii) Following mitosis, what must happen to stem cells, in order for them to repair the fractured bone? [1]

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(iii) State the name of the disease that can occur if cell division by mitosis is uncontrolled. [1]

.....

(c) Stem cells can be obtained from both adults and embryos.

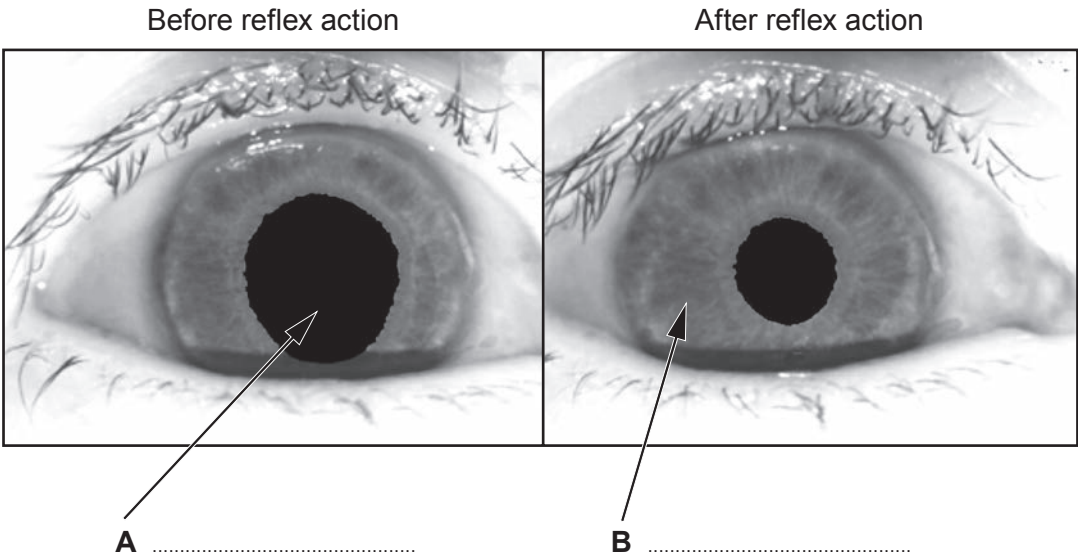
Give **one** reason why some people have a strong personal objection to the use of embryonic stem cells in medical research. [1]

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2. The photographs below show the appearance of an eye before and after a reflex action which occurs in response to a change in light intensity.



- (a) (i) Label **A** and **B** on the diagram. [1]
- (ii) From the photographs, describe how **and** why parts **A** and **B** of the eye alter when the light intensity changes. [3]

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- (iii) State **two** features of **all** reflex actions. [1]

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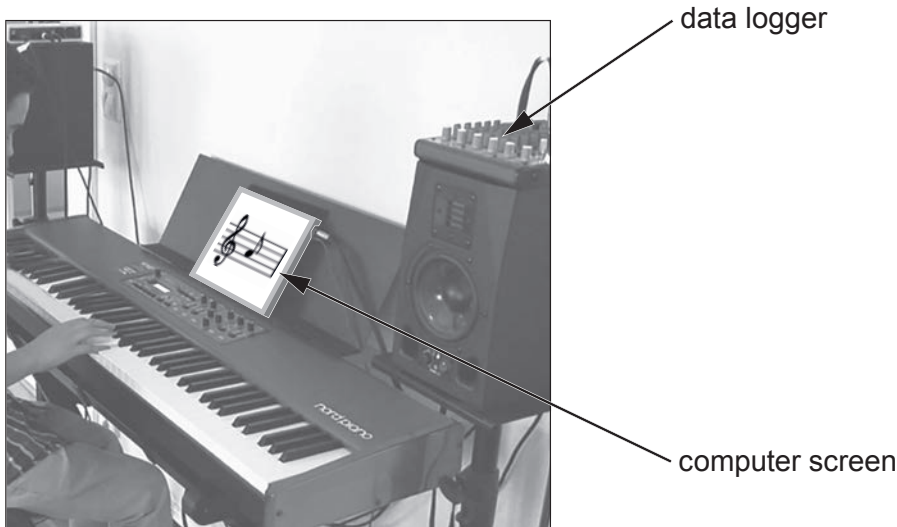


(b) Josie investigated reaction time in humans.

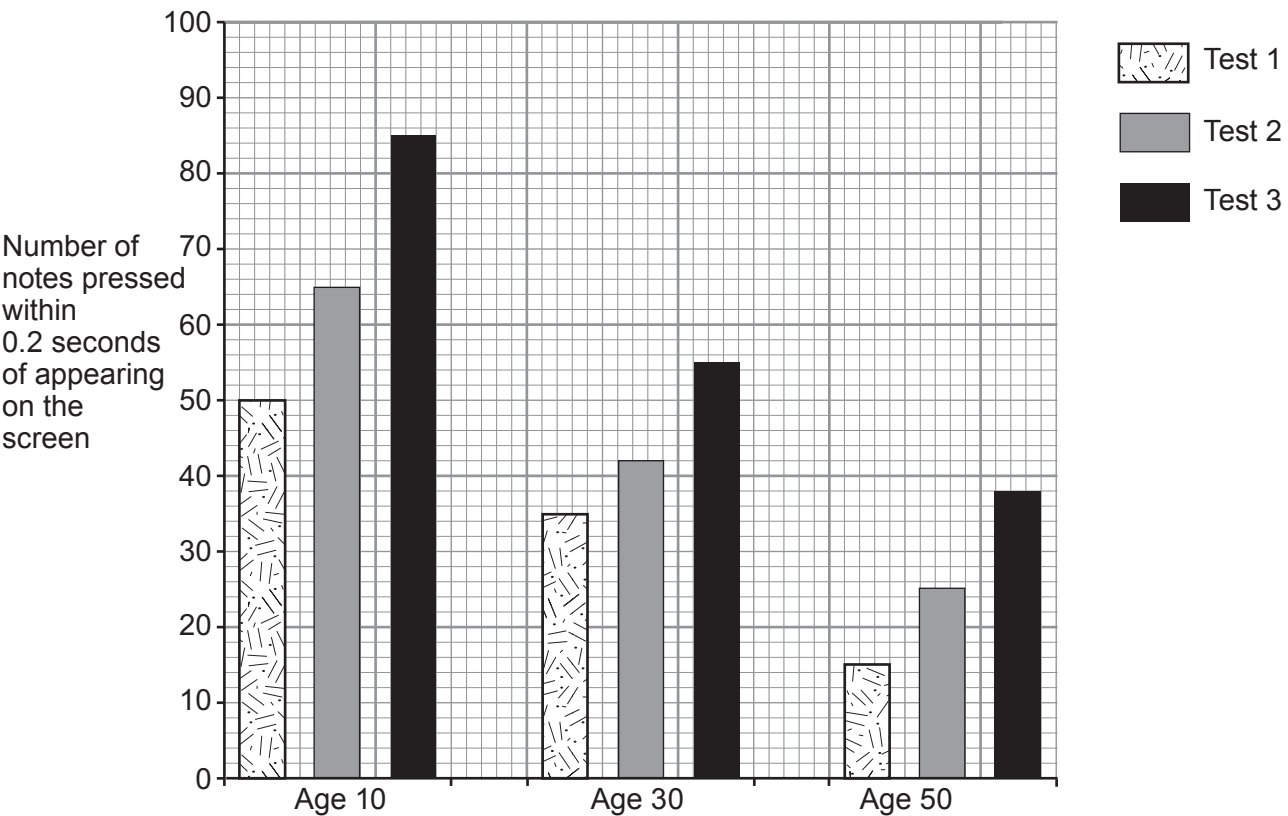
She tested three people, two males and one female of ages 10, 30 and 50 years old. They had between one and 20 years experience of playing the keyboard.

By means of a computer app, 90 random music notes flashed one by one onto a screen. The person being tested then instantly pressed each note on the keyboard as soon as it was seen. Each person did the test three times. No incorrect notes were pressed.

A data logger recorded the number of notes which were pressed within 0.2 seconds of appearing on the screen.



The bar chart shows the results of the investigation.



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Use the bar chart to answer the questions.

(i) How does repeating the test affect reaction time? [1]

.....

(ii) Calculate the percentage change between tests 1 and 2 for age 50. Give your answer to one decimal place. [2]

Percentage change = %

(iii) From the data, what **two** conclusions could you make about the effects of age on reaction time? [2]

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(iv) I. Josie decided to try the investigation again and make it a fairer test of the effects of age. State **two** variables which she should control. [2]

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II. State **one other** way in which the investigation could be improved. [1]

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3. Describe the structure and function of DNA (Diagrams will not be credited). [6 QER]

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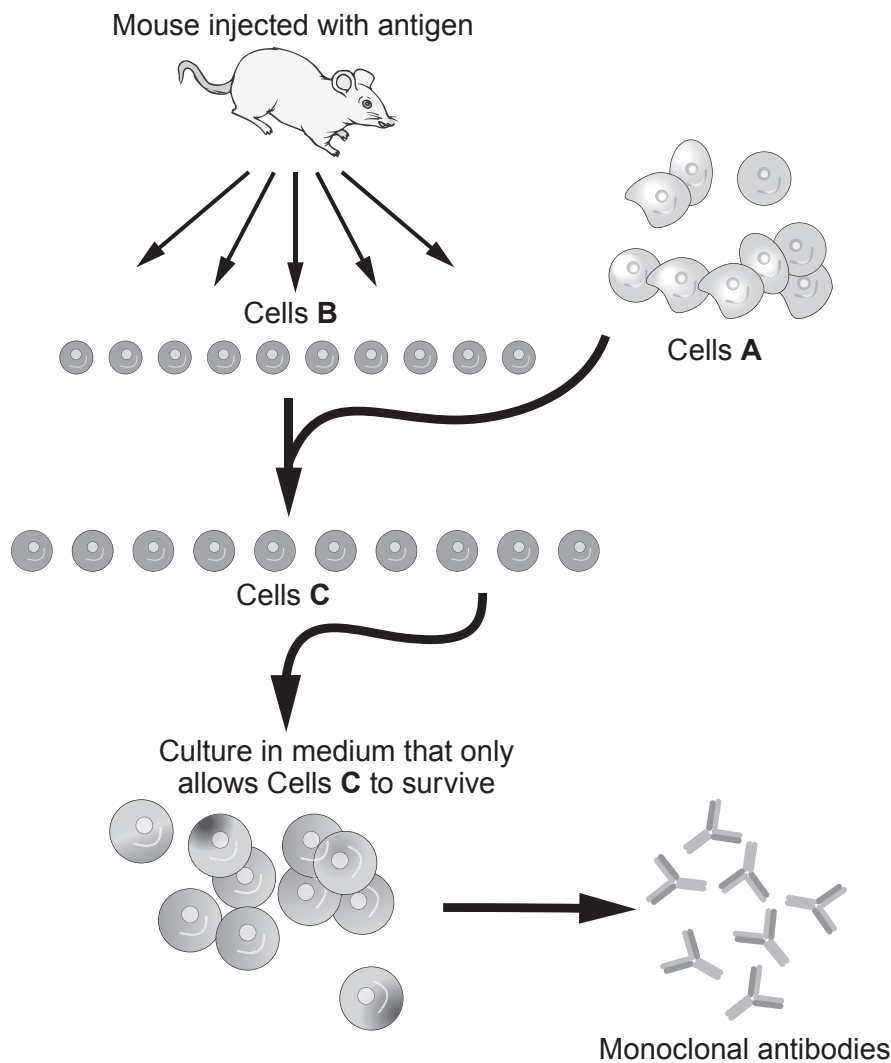


4. Basiliximab is a monoclonal antibody used to suppress the immune system of kidney transplant patients. It works by preventing white blood cells from attacking the cells of donor organs and therefore reduces the probability of rejection.

(a) (i) Apart from rejection, state another disadvantage of kidney transplants. [1]

(ii) Explain why and how white blood cells of the recipient would attack the cells of the donor organ therefore leading to rejection. [3]

(b) The diagram below shows a process for producing monoclonal antibodies.



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only

Use the diagram and your own knowledge to answer the following questions.

State the name of the cells labelled:

- (i) **A** [1]
- (ii) **B** [1]
- (iii) **C** [1]
- (iv) State how the following cells were produced: [2]

B
.....
C
.....

- (c) State **two other** examples of medical uses for monoclonal antibodies. [2]

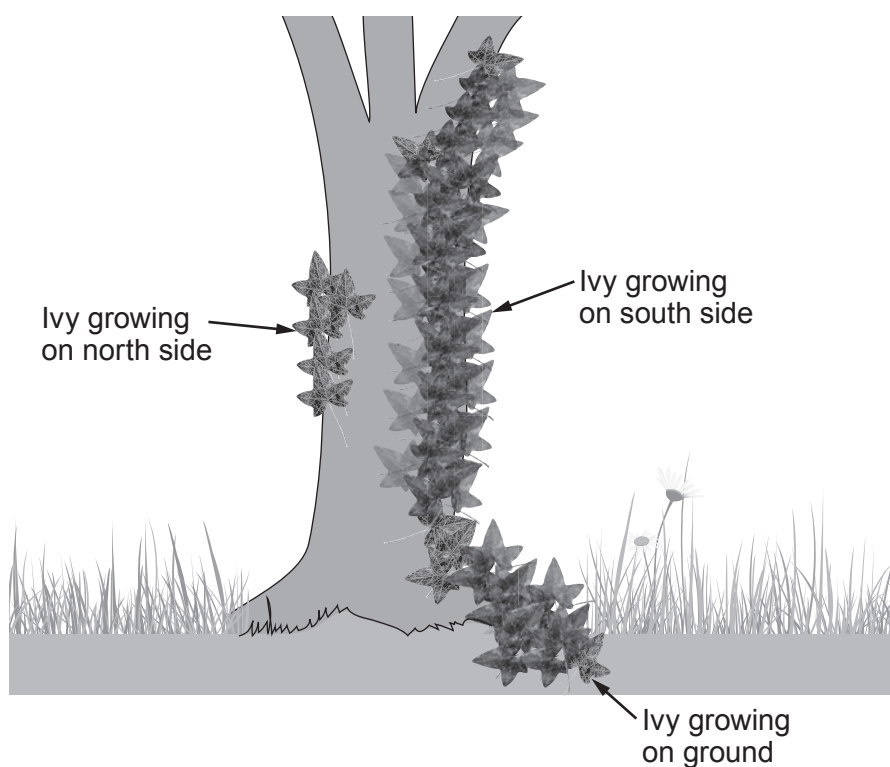
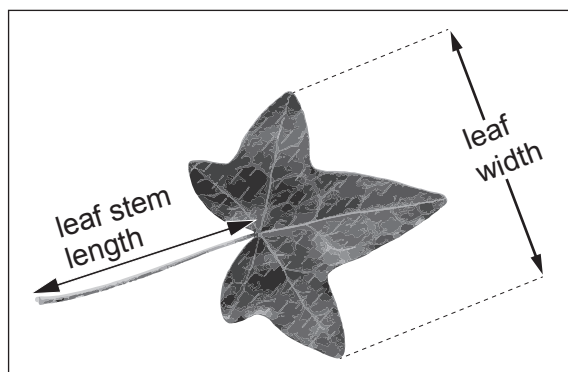
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5. Students investigated factors affecting leaf growth of ivy (*Hedera sp.*).

They measured the leaf stem length and the leaf width of 50 leaves of an ivy plant growing on the south side of a tree trunk at a height between one and two metres.



They compared these measurements with ivy leaves of the same ivy plant growing on the north side of the same tree trunk at the same height and also leaves from the same ivy plant growing flat on the ground.



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Their results are shown below:

	Leaves from ivy growing on the south side	Leaves from ivy growing on the north side	Leaves from ivy growing on the ground
Mean leaf stem length (mm)	63	60	42
Mean leaf width (mm)	55	52	38
Ratio of leaf stem length to leaf width	1.15 : 1	 :	1.11 : 1

(a) (i) Calculate the ratio of leaf stem length to leaf width for the ivy growing on the north side of the tree and **write your answer in the table.** [2]
SPACE FOR WORKING

(ii) Suggest why there is a difference between the ratio of leaf stem length to leaf width for the ivy growing on the ground and the ivy growing on the south side of the tree. [1]

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.....

(b) Students recorded the data for the ratio in the frequency table below.

Ratio	South facing	North facing	Ground growing
0.51 - 0.75	7	4	4
0.76 - 1.00	9	7	13
1.01 - 1.25	15	27	24
1.26 - 1.50	13	9	7
1.51 - 1.75	3	5	2
1.76 - 2.00	0	0	0

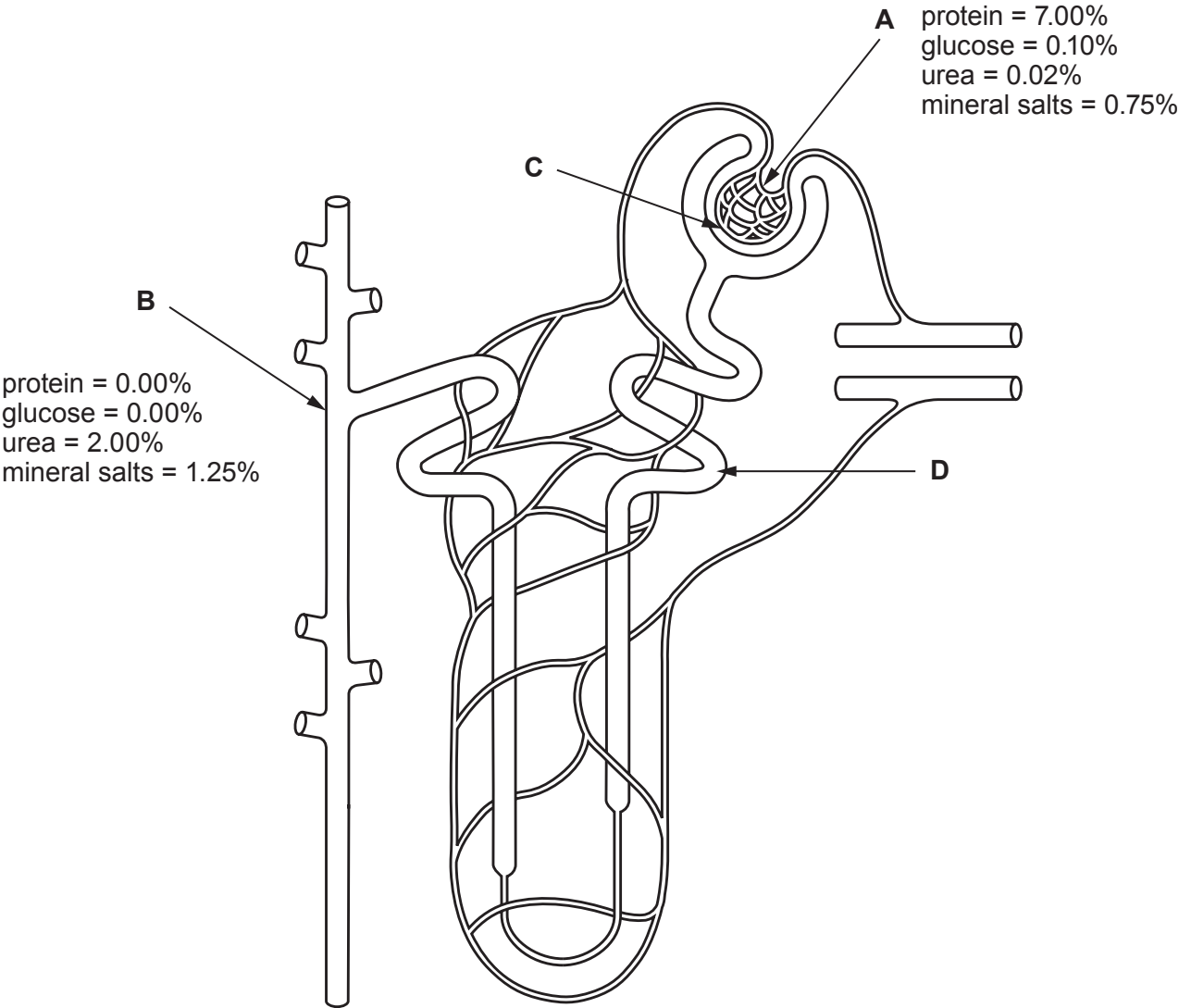
State the type of variation shown by the ratio. [1]

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6. The diagram below represents a human nephron. The concentration of various substances is shown at two different points, **A** and **B**.



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(a) Use the diagram and your own knowledge to answer the following questions.

(i) Name and describe the process occurring at point **C**. [2]

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.....

.....

(ii) Explain why the concentration of glucose is different at points **A** and **B**. [1]

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(iii) State the name of structure **D**. [1]

.....

(b) State a reason for the increase in the concentration of urea and mineral salts between points **A** and **B**. [1]

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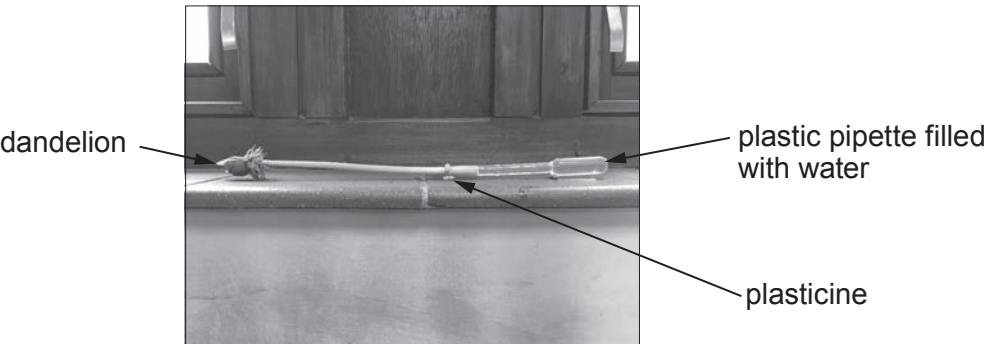
7. Some students investigated the tropic responses of the dandelion (*Taraxacum officinale*).

(a) State what is meant by tropic responses. [1]

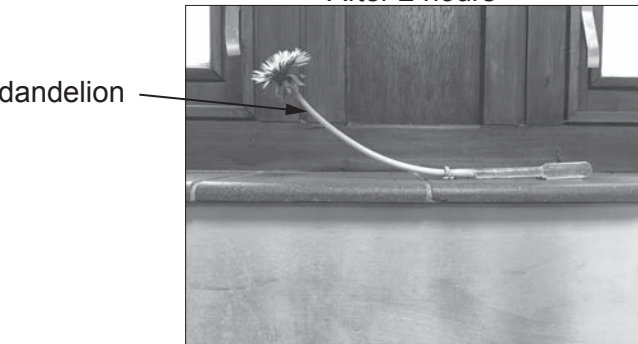
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(b) The students set up the apparatus shown below on a south-facing window-sill.



After 2 hours



The students concluded that the dandelion had shown positive phototropism.

Suggest why the students cannot be confident in their conclusion. Explain how they could change the method to improve confidence in their conclusion. [3]

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(c) State the name of the plant hormone responsible for phototropism. [1]

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8.

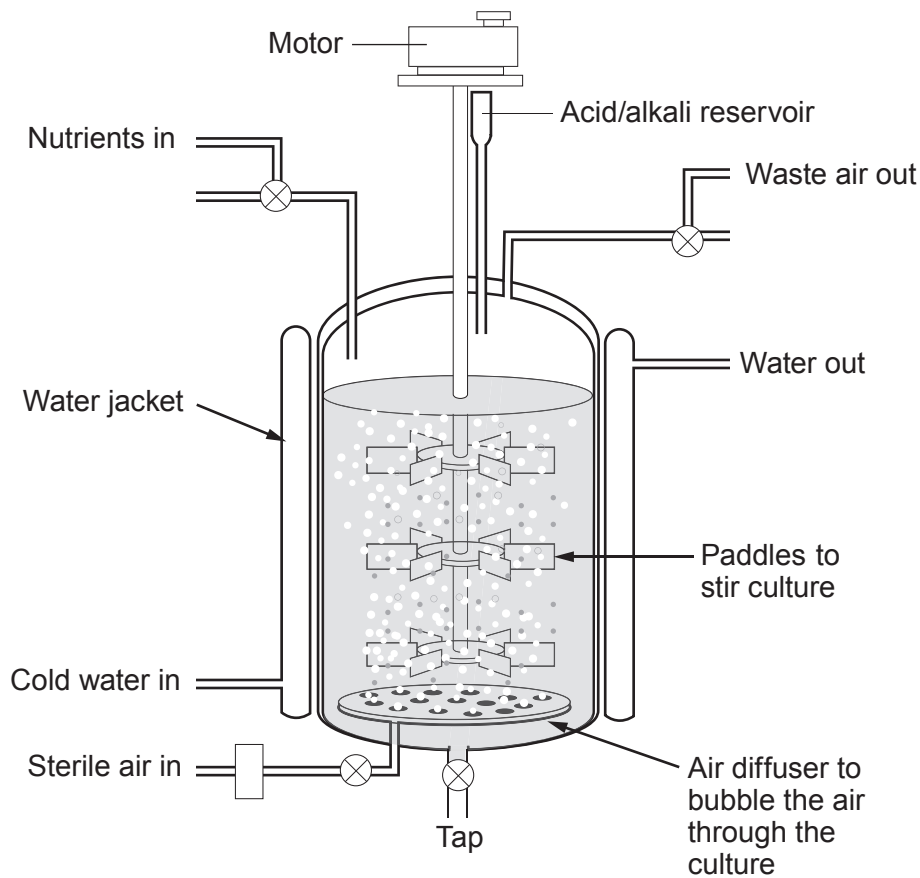
Thanks to PENICILLIN
... He will come home!



Penicillin is an antibiotic that is produced by several species of a fungus from the genus *Penicillium*. Its effect was first identified by Alexander Fleming in 1928. It was only in 1939 that Penicillin was extracted by Howard Florey and Ernst Chain at Oxford University and later developed for use as an antibiotic. The industrial scale production of penicillin was developed in the USA between 1941 and 1943 in response to the high number of soldiers who were dying due to diseases linked to their wounds during World War II.

Penicillium can be grown industrially in a tank called a fermenter. An example is shown in the diagram opposite:





(a) (i) State how penicillin helped save the lives of wounded soldiers. [1]

.....

.....

(ii) Name a group of microorganisms that would not be affected by antibiotics. [1]

.....

(iii) State a danger of new drug treatments. [1]

.....

(b) Use the diagram of the fermenter and your own knowledge to answer the following question.

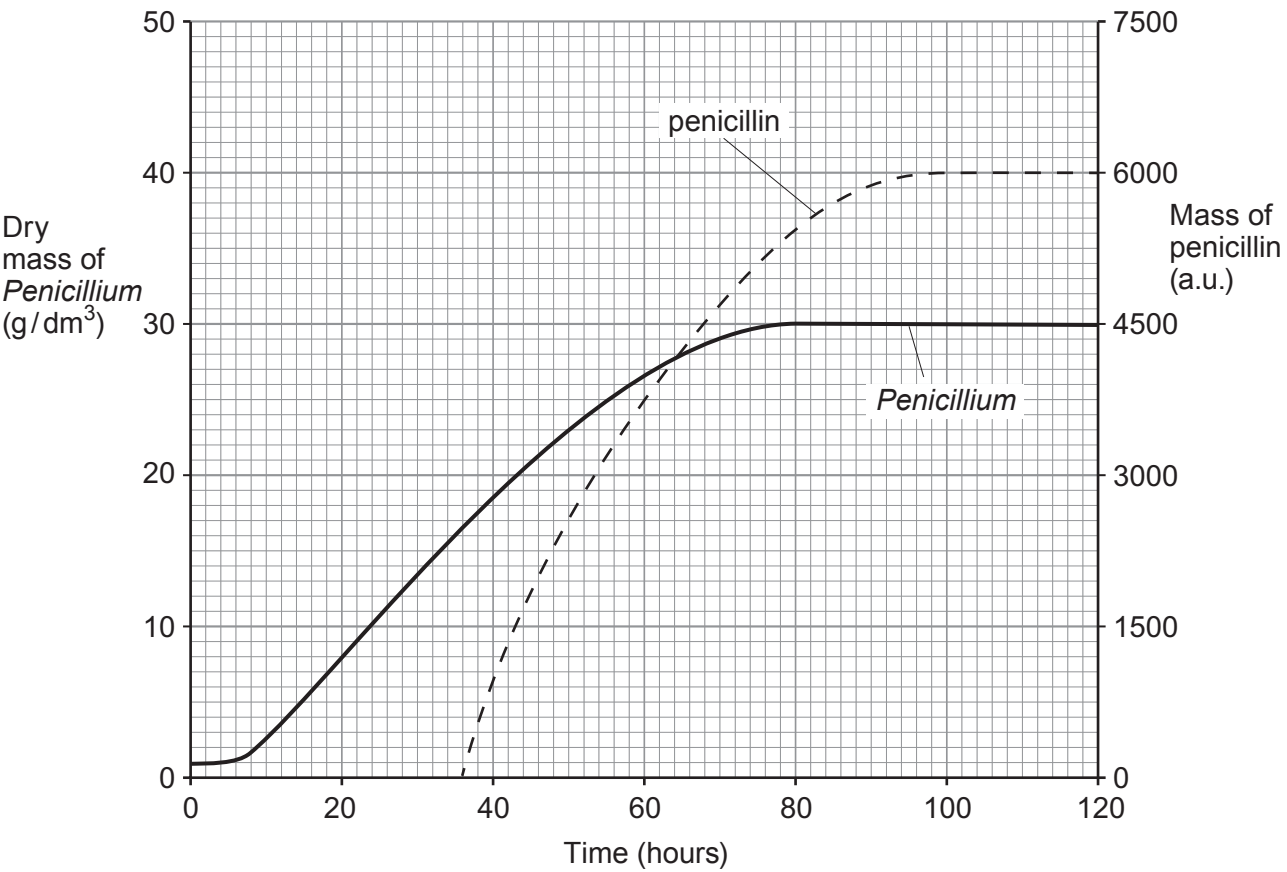
Other than nutrients, state **one** factor that affects the growth of *Penicillium* and state how it can be controlled by the fermenter. [1]

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(c) The graph below shows the quantity of penicillin and *Penicillium* inside the fermenter at 25°C over a period of time:



Using the graph:

- (i) Calculate the mean growth rate of *Penicillium* between 20 and 50 hours. [2]

Growth rate = g/dm³/hour



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(ii) Identify the optimum time for the penicillin to be extracted from the fermenter and justify your answer. [2]

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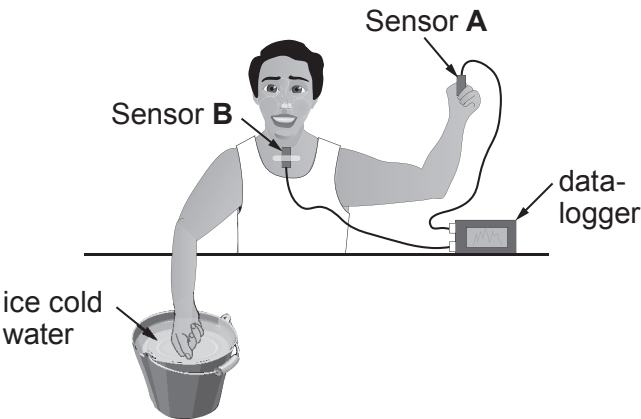
(d) Draw a line on the graph to estimate the effect of increasing temperature from 25°C to 28°C on the dry mass of *Penicillium* produced between 0-120 hours. [1]

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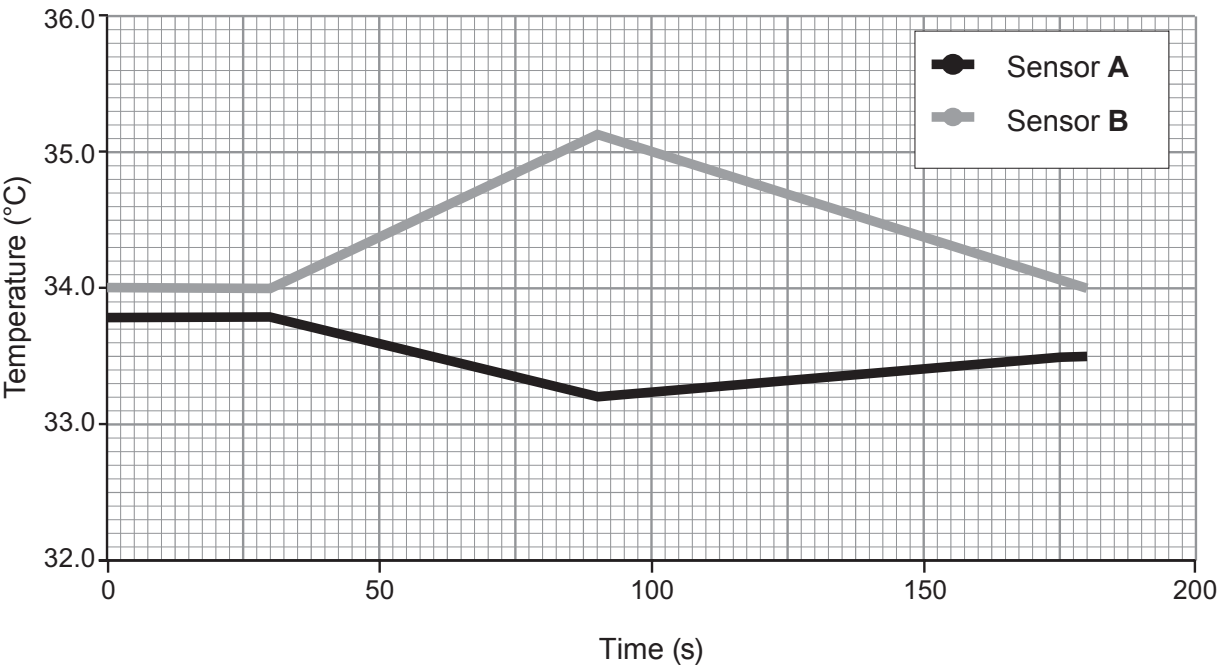
9. Dylan set up an experiment to study negative feedback mechanisms associated with rapid cooling of the hand.

A temperature sensor was held between the fingertips of a volunteer and another sensor was taped to the skin on the upper chest. The sensors were connected to a data logger that was set to record for three minutes.



The volunteer placed their right hand into ice cold water 30 seconds after the recording started. The hand was then removed from the ice cold water after a further 30 seconds.

The results from the data logger are shown below:



Examiner
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(a) State what is meant by a negative feedback mechanism. [2]

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(b) (i) Describe the results for sensor **A** after the hand was placed in the ice cold water. [1]

(ii) Explain the change in temperature recorded by sensor **A** between 30 seconds and 90 seconds. [4]

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(c) Suggest the reason for the increase in the temperature recorded by sensor **B** between 30 seconds and 90 seconds. [1]

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(d) Suggest how the results of the experiment would differ if the volunteer had consumed an alcoholic drink 20 minutes before the start of the experiment. [1]

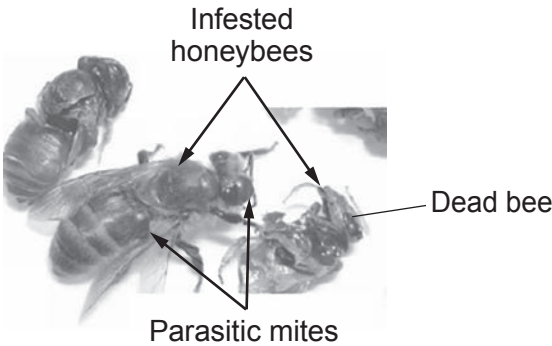
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10. *Varroa destructor* is a parasitic mite of honeybees. A colony of honeybees affected by the parasite becomes weakened and eventually dies out. The parasitic mite originated in Asia, where it is a pest of the Asian honeybee, *Apis cerana*. It has spread across most continents, arriving in the UK in the early 1990s and affecting the native European honeybee, *Apis mellifera*.



- (a) Use your knowledge of classification to suggest why the parasitic mite was able to spread from the Asian to the European honeybee. [2]

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In 2014-2015 there were 1650 registered beekeepers in Wales who kept bee colonies, each of which is one beehive. The National Bee Unit recommends that honeybees should be treated with pesticides to try and reduce the numbers of the parasitic mite in beehives.

In recent years many beekeepers in North West Wales have stopped treating their bees with pesticides. These beekeepers believe that there is no difference in colony losses between bees treated with pesticides and bees not treated with pesticide.



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The table below shows Winter Losses (the number of colonies that died out) between 2010-2015 in North West Wales.

Season	Number of beekeepers in the survey		Total number of colonies	Number of treated colonies	% winter loss in treated colonies	Number of colonies not treated	% winter loss in not treated colonies
	Treating	Not treating					
2010-2011	10	5	71	44	27	27	11
2011-2012	11	31	355	180	8	175	7
2012-2013	8	46	251	75	41	176	32
2013-2014	12	55	396	81	9	315	6
2014-2015	17	65	500	97	8	403	8

(b) What percentage of beekeepers in Wales took part in this survey in 2014-2015? [2]

percentage of beekeepers = %

(c) (i) Use the information in the table to state **one** conclusion about the effect of the pesticides on winter losses of bees. [1]

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Examiner
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(ii) Give **two** reasons why some beekeepers across the UK would argue that the strength of evidence in this survey is weak. [2]

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(d) Early attempts at control of the parasitic mite involved using the chemical, pyrethroid. By the late 1990s the parasitic mite developed resistance to the chemical and beekeepers were advised to use other chemicals to control the pest.

Explain why, over time, the pyrethroid chemical became less effective at killing the parasitic mite. [4]

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



GCSE

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BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

HIGHER TIER

TUESDAY, 14 MAY 2019 – AFTERNOON
1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	10	
3.	6	
4.	9	
5.	10	
6.	8	
7.	6	
8.	13	
9.	8	
Total	80	

ADDITIONAL MATERIALS

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INFORMATION FOR CANDIDATES

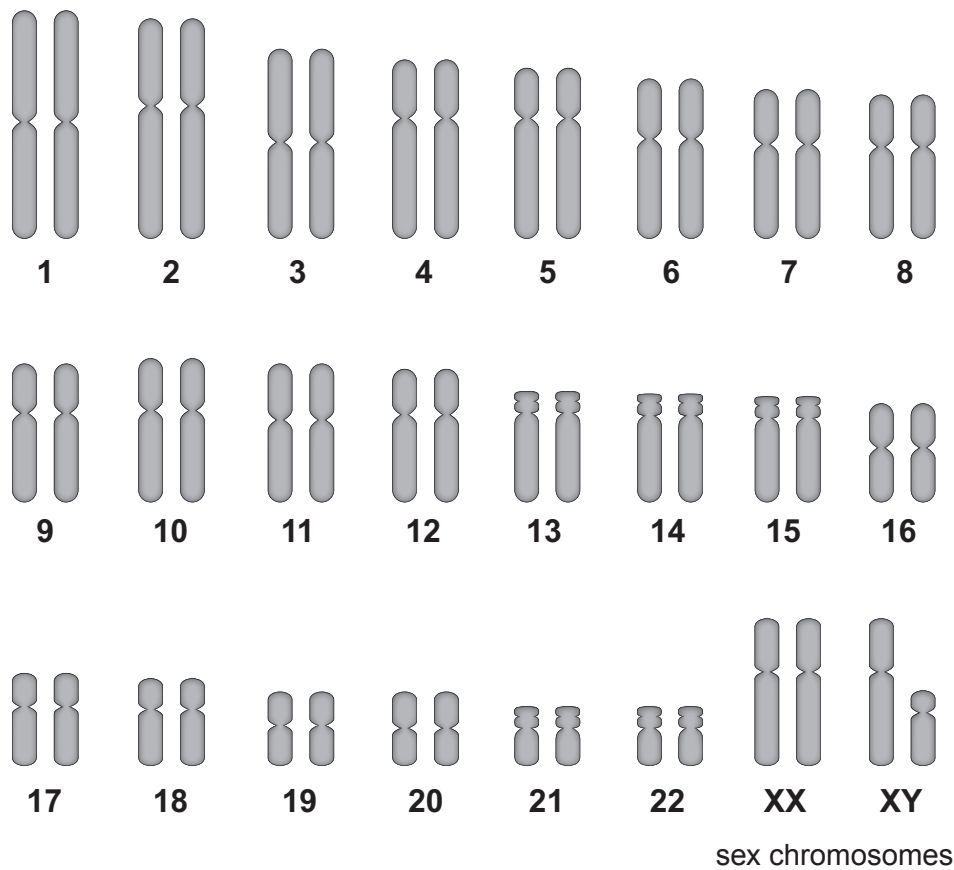
The number of marks is given in brackets at the end of each question or part-question.
Question **6(b)** is a quality of extended response (QER) question where your writing skills will be assessed.



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Answer all questions.

1. (a) The image shows human chromosomes.



- (i) State the number of pairs of chromosomes present in a single body cell of a human. [1]
.....
- (ii) State which sex chromosomes are present: [1]
in body cells of a man;
in body cells of a woman.



(iii) Complete the Punnett square below to show the sex chromosomes in the gametes of a male and female parent and in their possible offspring. [2]

		Male Parent	
Female Parent	Gametes		

(b) Height in humans is controlled by many genes as well as by environmental factors. Females usually reach their full height by age 18 and males by 25.

Students investigated the heights of male and female airline cabin crew. All cabin crew must be between 1.58 m and 1.90 m in height and between 18 and 45 years of age.

The students said that they expected the males to be taller than the females.

They collected their data by selecting seven males and seven females at random and asking them to state their heights. They gave their heights in feet and inches and the students converted the data into metres.

The results, expressed to two decimal places, are shown in the table below.

Heights of females (m)	Heights of males (m)
1.80	1.84
1.78	1.80
1.83	1.72
1.68	1.70
1.75	1.61
1.82	1.81
1.69	1.73
mean height =	mean height = 1.74



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(i) **Complete the table** by calculating the mean height for females. [2]
Space for working.

(ii) I. State the hypothesis that the students were testing in their investigation. [1]

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II. State whether the results of the investigation support their hypothesis, giving the reason for your answer. [1]

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III. Give **one** way in which the strength of the evidence could be improved. [1]

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(c) State **one** source of inaccuracy in the method. [1]

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2. Cystic Fibrosis (CF) is a serious medical condition which affects the lungs. It results from a mutation in the DNA of a single gene which gives rise to a recessive allele.

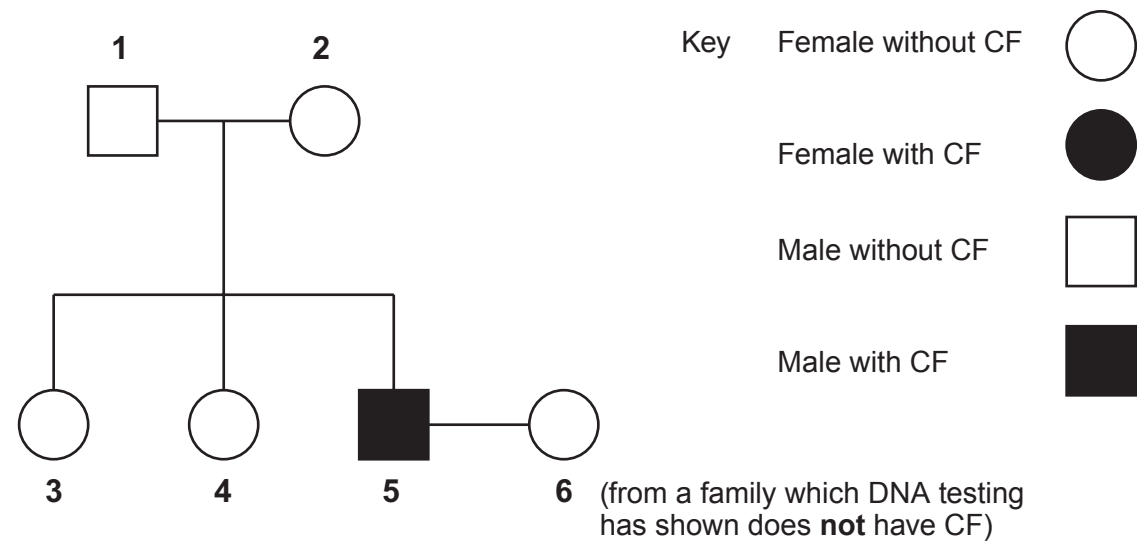
(a) Explain the meaning of the term *recessive allele*. [2]

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(b) The diagram below shows part of the family tree of a family which has one member who has CF.



(i) State the effect of a mutation on DNA. [1]

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.....

(ii) From the family tree:

I. State the numbers of **two** individuals who are known to be heterozygous for the allele which causes CF. [1]

.....

II. Suggest what advice a genetic counsellor would give individuals **5** and **6** about the chance of them having a child with CF. Explain your answer. [2]

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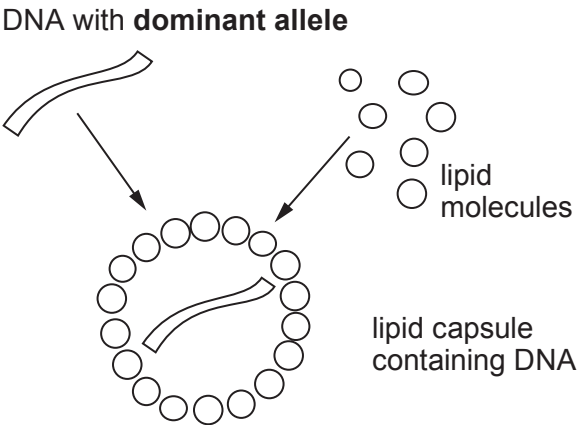
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- (c) CF is usually treated by intensive physiotherapy which can be very stressful. Medical scientists are now working to develop an alternative treatment called *gene therapy*. This process is summarised in the flow chart below.



Physiotherapy for CF



Gene therapy for CF

- (i) Lipid capsules are introduced into the lungs. State how this would be done. [1]

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- (ii) In order for this therapy to be successful two further difficulties must be overcome. Describe these **two** difficulties and suggest why this therapy is **not** a permanent cure. [3]

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Examiner
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3. (a) (i) State what is meant by a sense organ. [1]

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(ii) Name the **two** components of the central nervous system. [1]

.....

(iii) State **three** properties of a reflex action. [1]

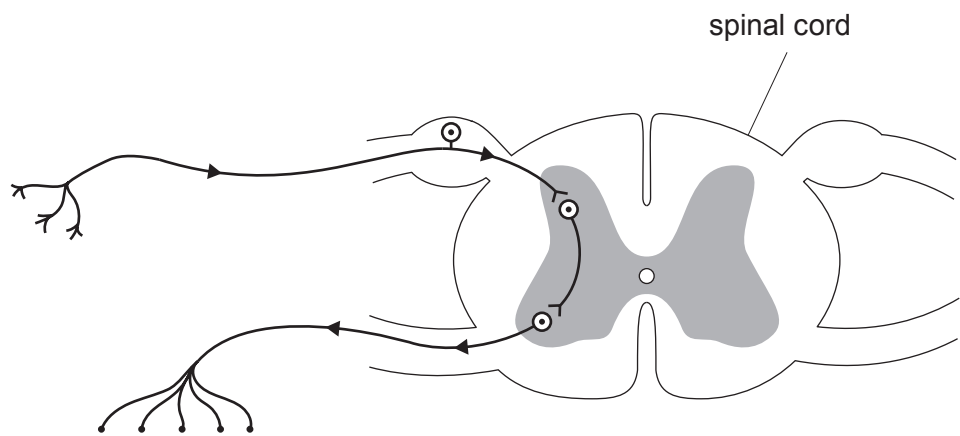
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(b) The diagram below shows a reflex arc. **On the diagram label:** [2]

- (i) the receptor
- (ii) the motor neurone



(c) Motor neurone disease is a rare condition that is caused by the motor neurones not functioning correctly. Suggest the effect of motor neurone disease on a reflex action. [1]

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6



4. Chlamydia is the most common sexually transmitted disease (STD) diagnosed in Wales.



In 2013, there were 5 076 reported cases of chlamydia. By 2014 this number had risen to 5 452, of which 71.1% were 16 to 24-year-olds.

(a) (i) Give the scientific name of the pathogen that causes the disease chlamydia. [1]

.....

(ii) State the name of the group of micro-organisms to which this pathogen belongs. [1]

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(b) Describe how the spread of chlamydia can be prevented. [1]

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(c) (i) Calculate the number of 16 to 24-year-olds diagnosed with chlamydia in 2014. [2]

Number of young people =



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(ii) Calculate the percentage increase in new chlamydia diagnoses between 2013 and 2014. [2]

Percentage increase =

(d) Suggest why public health awareness campaigns targeting the spread of chlamydia are important. [2]

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5. There are many different species of snail. Students were asked to estimate the population density of the garden snail (*Cornu aspersum*) in a local park using the following method.



Method

- Use a tape measure to mark out an area of 30 m × 30 m.
- Search for snails in the marked area and pick them off any plants or the ground and place them in a bucket.
- Use an identification key to identify the garden snails.
- Count the number of garden snails caught and mark their shells with nail varnish.
- Allow the nail varnish to dry before releasing the garden snails.
- Search the marked area again for snails after 1 week.
- Count the total number of garden snails collected and the number of marked garden snails which you have found.
- Release the snails.
- Estimate the size of the population of garden snails for the area sampled using the formula:

$$P = \frac{a \times b}{c}$$

where P = size of population in sample area

a = number of garden snails collected, marked and released in the first sample

b = total number of garden snails collected in second sample

c = number of marked garden snails collected in second sample

Results

- Number of garden snails in first sample = 198
- Total number of garden snails in second sample = 152
- Number of marked garden snails in second sample = 8



Examiner
only

(a) (i) Garden snails are invertebrates. State what is meant by the term *invertebrate*. [1]

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.....

(ii) Explain why it is important to use a scientific name to identify and classify organisms. [1]

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(b) (i) Use the formula given opposite to estimate the population of garden snails in the area sampled. [2]

Population of garden snails =

(ii) Use your answer from part (b)(i) to calculate the population density per m² of the garden snails. [2]

Population density = snails per m²

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Examiner
only

(c) State **two** assumptions made when using the method given on page 12 to estimate the population density of garden snails. [2]

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(d) Suggest how the students could improve the strength of evidence in their survey. [1]

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(e) Suggest **two** precautions that the students should take during this survey to reduce the risk of harm to individual snails or their habitat. [1]

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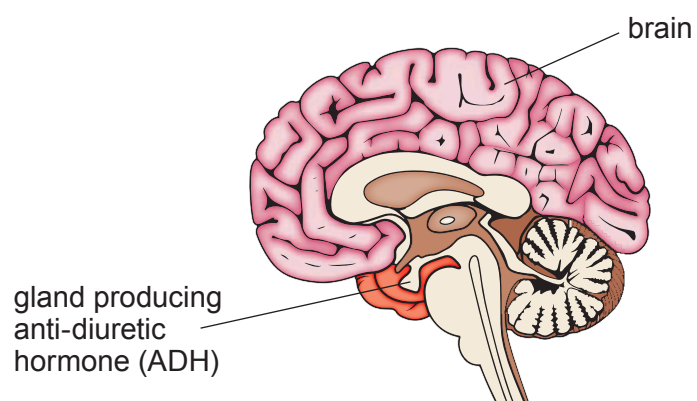
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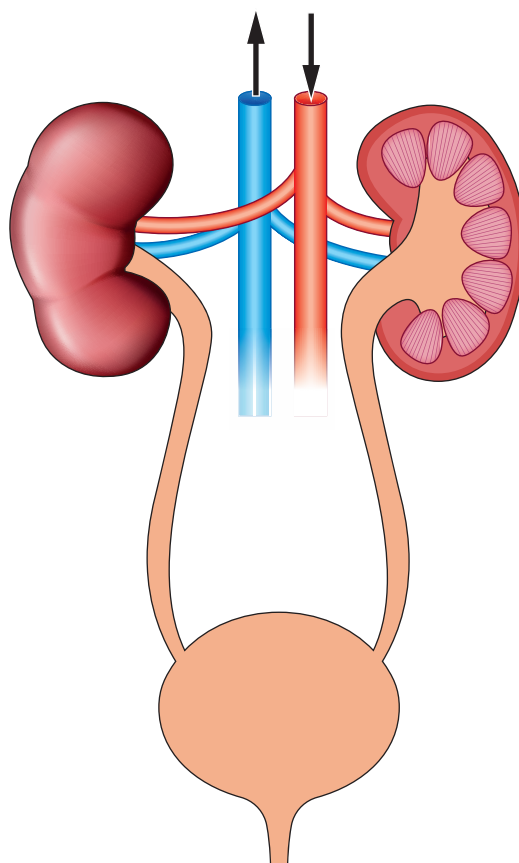


6. The diagrams below show a section through the human brain and also the excretory system.

The brain



The excretory system



- (a) On the diagram of the excretory system, label:

[2]

- (i) the aorta;
- (ii) the medulla of a kidney.



Examiner
only

(b) Explain the link between the quantity of anti-diuretic hormone (ADH) produced by the brain and the volume and concentration of urine produced in a person who is dehydrated.
[6 QER]

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7. The article below was taken from a newspaper in early 2017.

“DNA profiling snares Welsh sheep rustler after investigation.”

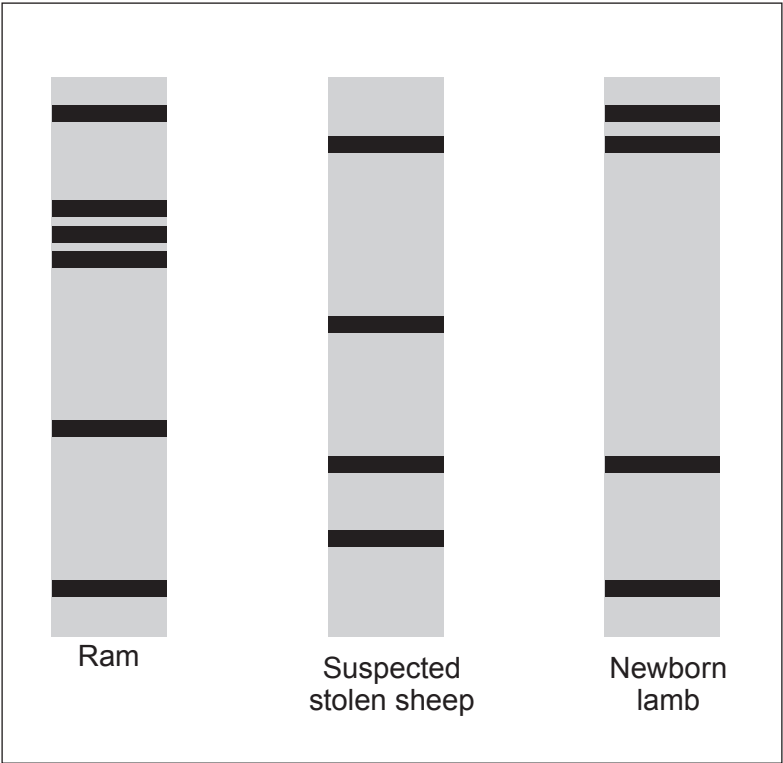
Daily Post 4/4/2017

In January 2015, a number of pregnant sheep were stolen from a farm. Enquiries within the local farming community revealed a pregnant sheep, which was suspected of having been stolen, had been sold at a livestock market.

The Animal Plant Health Agency (APHA) wanted to carry out DNA analysis on the sheep and its lamb. They had to wait two months for the sheep to give birth before taking a sample of blood from the lamb.

The diagram below shows a genetic profile of:

- the ram (known to be the father of the lamb in the suspected stolen sheep)
- the suspected stolen sheep
- the newborn lamb.



Examiner
only

(a) Describe the process of genetic profiling. [2]

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(b) Explain how the genetic profile provides evidence that the sheep had been stolen. [2]

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(c) State **one** possible health benefit and suggest **one** ethical concern linked to DNA profiling in humans. [2]

Possible health benefit

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Ethical concern

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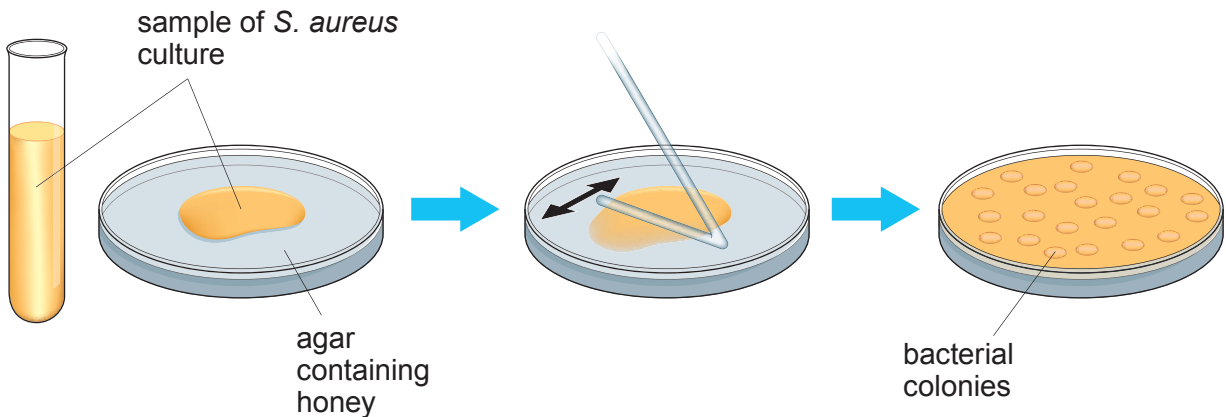
8. *Staphylococcus aureus* is a bacterium which is a common cause of infection in human wounds.

S. aureus has become resistant to antibiotics.

Honey has been used as a wound dressing for thousands of years. Scientists carried out a laboratory investigation to study the effect of honey on the growth of *S. aureus*.

A series of agar plates containing different honey concentrations was prepared. A culture containing *S. aureus* was transferred to each of the agar plates.

- 1 0.3 mm³ of *S. aureus* culture poured onto honey agar
- 2 Spread sample evenly over the surface
- 3 Agar containing honey at 3% concentration at 24 hours.



The plates were incubated for 24 hours and then inspected for the presence of bacterial colonies. The results are shown below.

Concentration of Honey (%)	Bacterial colonies present (✓) or absent (✗)
0	✓
1	✓
2	✓
3	✓
4	✗
5	✗
6	✗
7	✗
8	✗
9	✗
10	✗



Examiner
only

- (a) State the link between the number of bacterial colonies present on the agar in diagram 3 at 24 hours and the number of bacteria in the original sample. [1]

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- (b) (i) Suggest the temperature at which the agar plates would have to be incubated in this investigation. Give a reason for your answer. [2]

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- (ii) State **two** basic aseptic techniques that should have been followed in preparing the plates shown in the diagram. [2]

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- (c) Using the diagrams, calculate the number of bacteria present in **1 cm³** of the original *S. aureus* culture. **Present your answer in standard form.**
(Note 1 cm³ = 1000 mm³) [3]

Number of bacteria present = bacteria / cm³



Examiner
only

- (d) (i) Using the table on page 20, state the minimum concentration of honey required to stop the growth of bacteria for 24 hours. [1]

Minimum concentration = %

- (ii) Suggest how scientists could develop their investigation to get a more accurate value for the minimum concentration of honey required to stop the growth of bacteria for 24 hours. [1]

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- (e) State the **three** pre-clinical stages that would have to be carried out in order to develop honey as a new medicine for the treatment of wounds. [3]

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13



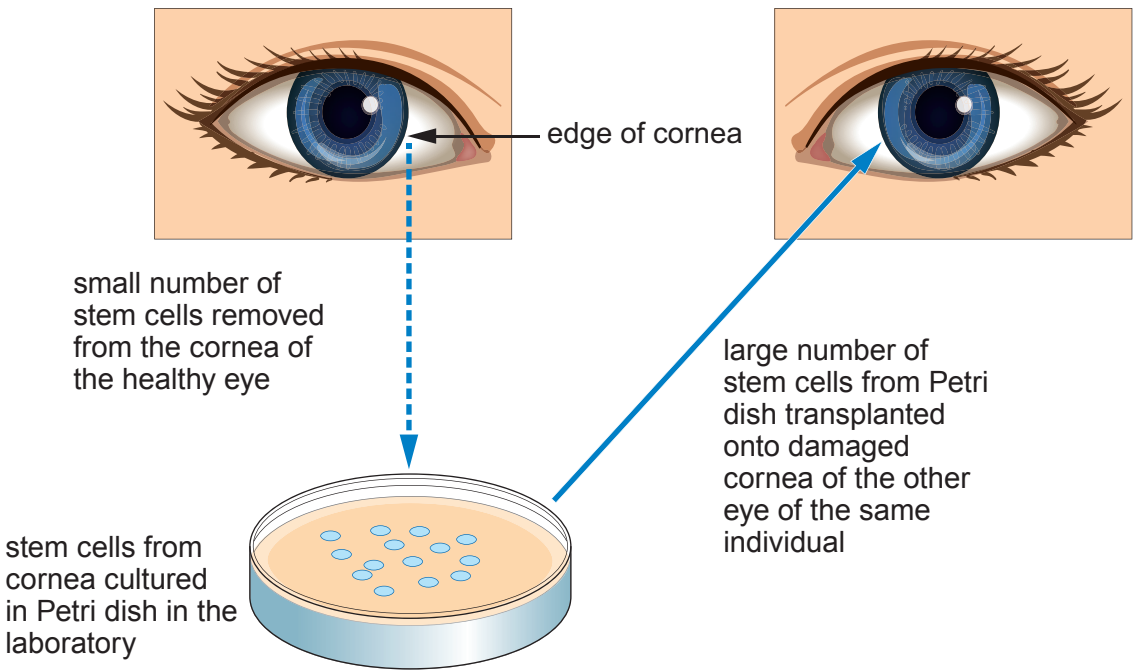
9. The eye is a complex organ made up of many tissues. Diseases of the eye occur when one or more of these tissues are damaged or stop functioning. Stem cell treatments are now being developed and used to treat many eye diseases.

Example 1 – Repairing the cornea (This treatment is now licensed for use.)

Stem cells at the edge of the cornea replace cells that are constantly being worn away. These stem cells may also be damaged by burns, radiation, genetic disorders, infection and drug use.

The diagram below shows a stem cell treatment to repair the cornea. This treatment uses stem cells from a healthy part of the cornea of an affected individual.

Diagram 1

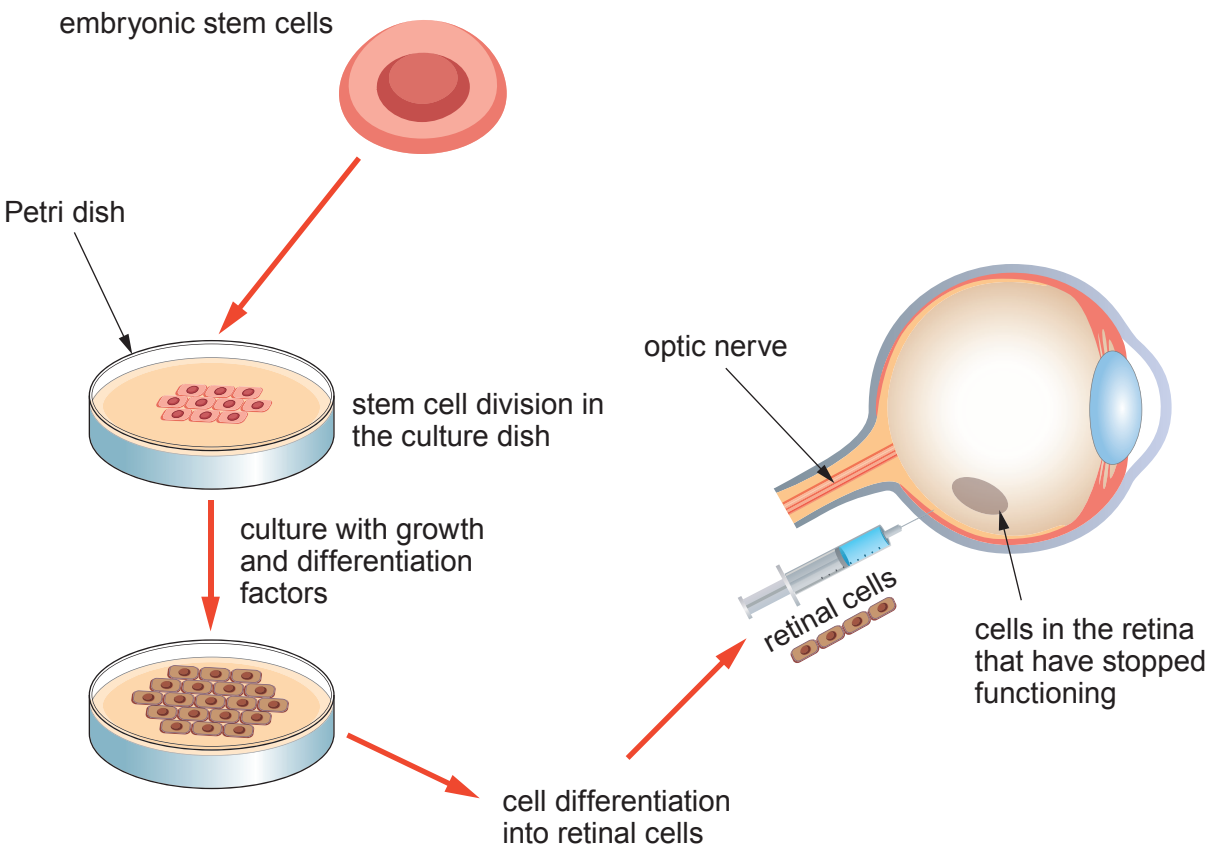


Example 2 – Repairing the Retina (This treatment is currently being researched and tested.)

In age-related macular degeneration (ARMD), cells in the retina stop functioning.

The diagram below shows a stem cell treatment for ARMD that uses embryonic stem cells to replace non-functioning cells in the retina.

Diagram 2



(a) State the function of the cornea. [1]

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.....

(b) Explain the meaning of the term *stem cell*. [1]

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Examiner
only

(c) State the name of the type of cell division that occurs in the Petri dishes in diagrams **1** and **2** and explain its significance in the treatments. [3]

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(d) Suggest why example **1** does not work if both eyes are badly damaged. [1]

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(e) State **two** advantages of using stem cells as a treatment in example **1** compared to their use in example **2**. [2]

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8

END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE – CONTINGENCY

3400UD0-1



WEDNESDAY, 22 JUNE 2022 – AFTERNOON

BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

HIGHER TIER
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	10	
3.	4	
4.	6	
5.	4	
6.	5	
7.	9	
8.	11	
9.	7	
10.	14	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the questions correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **10(c)(iii)** is a quality of extended response (QER) question where your writing skills will be assessed.

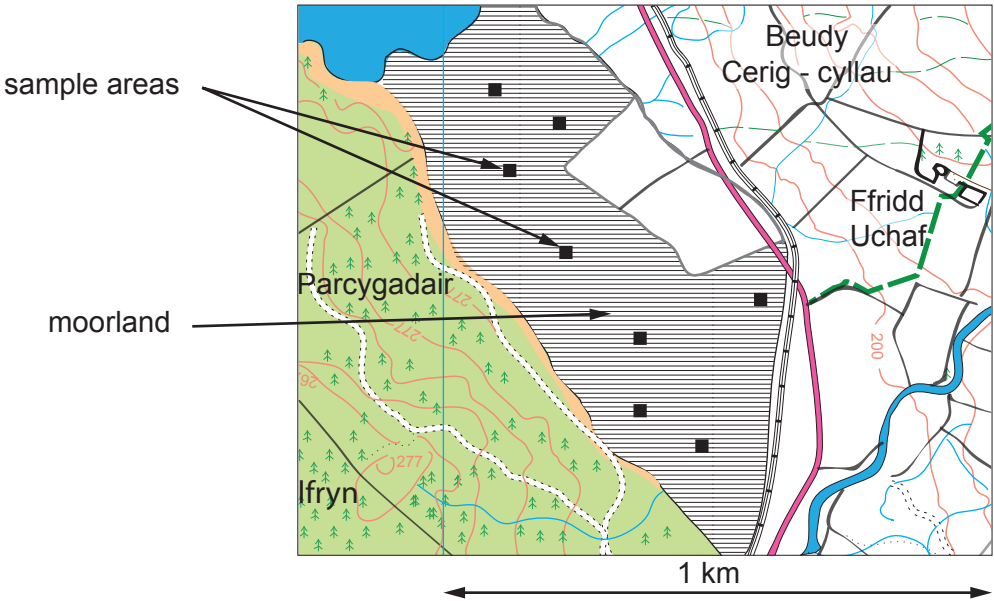


JUN223400UD0101

Answer **all** questions.

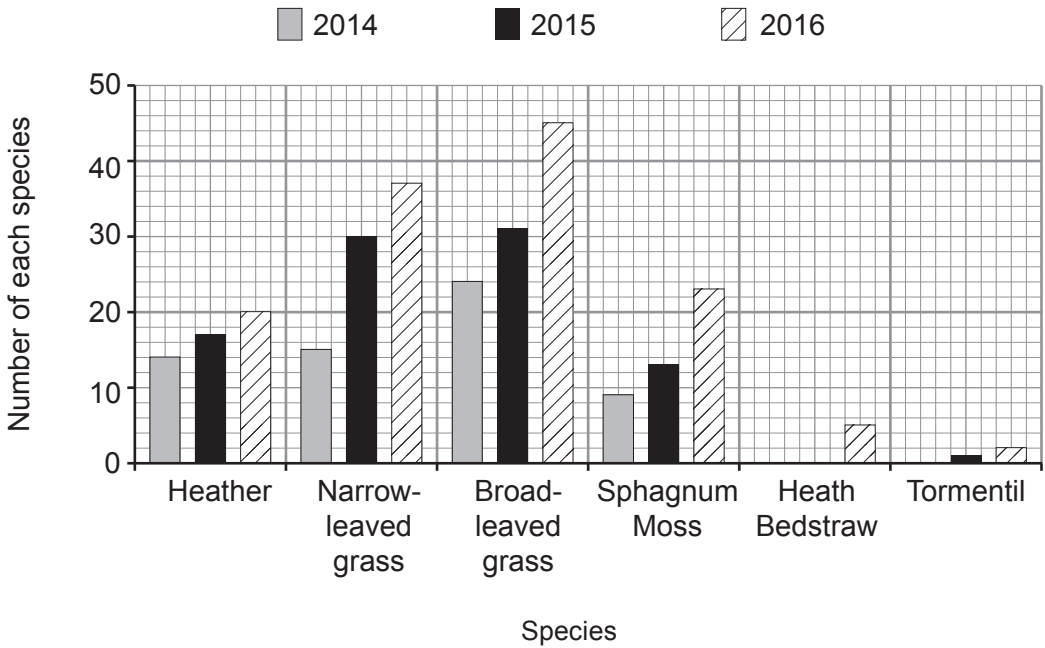
1. A group of students investigated the biodiversity of a moorland over a period of three years. Two years before the investigation started, the drainage ditches in the moorland had been closed in order to increase the water content of the land. **Image 1.1** is a map showing the moorland (shaded area) being investigated. The black squares represent locations of eight sample areas. Each sample area is 625 m² (25 m × 25 m).

Image 1.1



The results of the survey are shown in **Graph 1.2**.

Graph 1.2



Examiner
only

(a) Describe the method that the students would have used to investigate the abundance of plant species in the moorland. [4]

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(b) (i) Calculate the percentage change in narrow-leaved grass between 2014 and 2016. [2]

Percentage change = %

(ii) State **two** pieces of evidence that show that the biodiversity of the moorland increased between 2014 and 2016. [2]

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(c) (i) Suggest why the students sampled several areas over the whole moorland. [1]

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(ii) Suggest how, over the three years, the students made sure the investigation was a fair test. [1]

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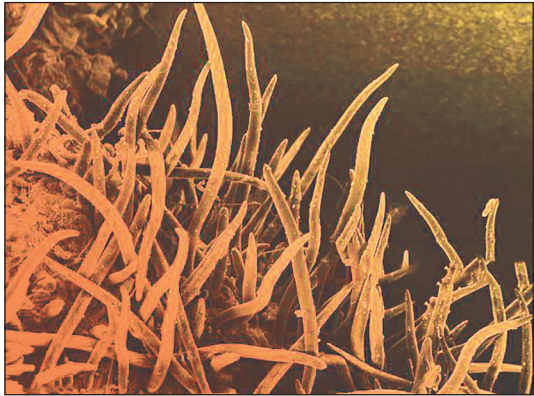
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10



2. A peach is a fruit which has tiny hairs on its surface. A nectarine is a type of peach where the hairs are absent. The production of hair on the surface is controlled by a single gene. A dominant allele codes for the production of hair. A mutation to this gene produces a recessive allele which does not produce hair. **Image 2.1** is a picture produced by an electron microscope showing the hair growing on the surface of a peach.

Image 2.1



Peach scab disease is caused by the fungus *Cladosporium carpophilum* and can affect both peaches and nectarines. Fungi reproduce by producing spores, which are spread by wind and rain. When the spore lands on the surface of a fruit, the fungus begins to grow and scabs soon develop. Nectarines are more likely than peaches to develop scab disease. **Image 2.2** is a picture of a nectarine showing scabs caused by *Cladosporium carpophilum*.

Image 2.2

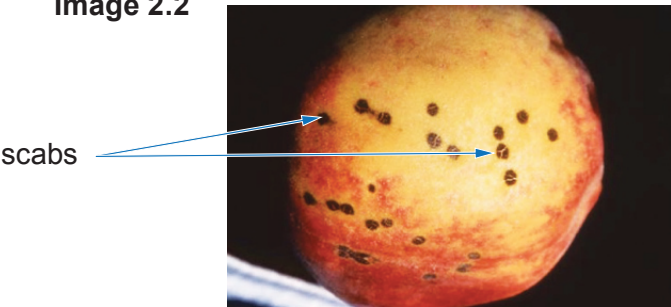
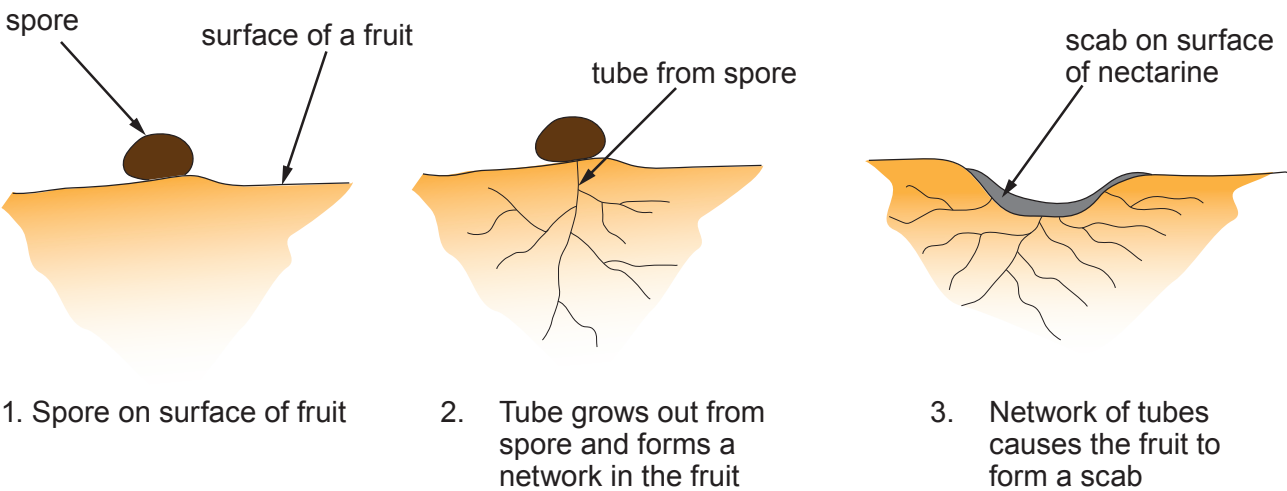


Image 2.3 shows a fungal spore developing and forming a scab.

Image 2.3



Examiner
only

(a) (i) State why scientists use the name *Cladosporium carpophilum* rather than the common name peach scab when discussing this disease. [1]

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(ii) Using all the information given, suggest why nectarines are more likely than peaches to develop scab disease. [2]

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(b) (i) State what is meant by the term allele. [1]

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(ii) State what is meant by a mutation. [1]

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(iii) Give **one** example of an **environmental** factor which increases the rate of mutations. [1]

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Continued overleaf

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Examiner
only

- (c) (i) Use the letters **H** and **h** to show the result of a genetic cross between two peach trees that would result in the production of seeds that could grow into nectarine trees. [3]

Key: **H** = allele for hairy fruit
h = allele for hairless fruit

Phenotype = peach tree × peach tree

Genotype = ×

Gametes		

- (ii) Using your answer from the Punnett square above, state the probability of producing a nectarine tree from this cross. [1]

Probability =

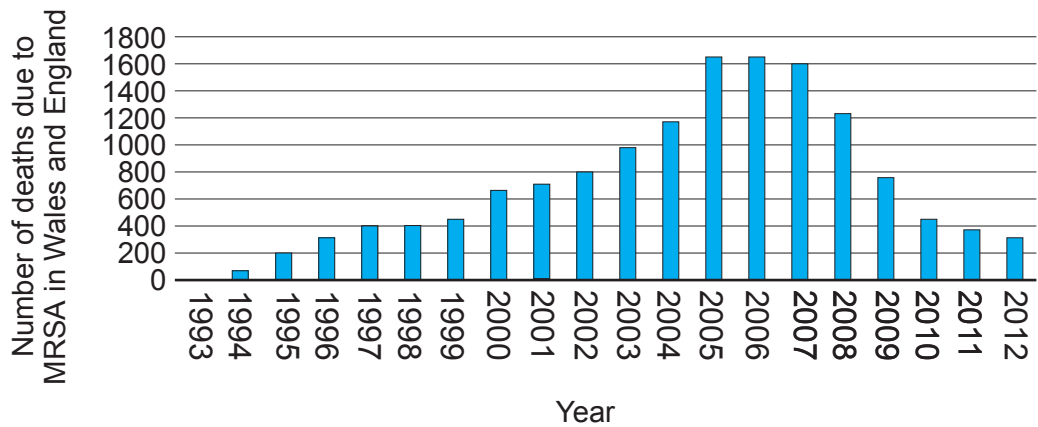
10



Examiner
only

3. Methicillin is a type of antibiotic which has often been used to treat infections. **Graph 3.1** shows the number of deaths in Wales and England from methicillin resistant *Staphylococcus aureus* (MRSA) between 1993 and 2012.

Graph 3.1



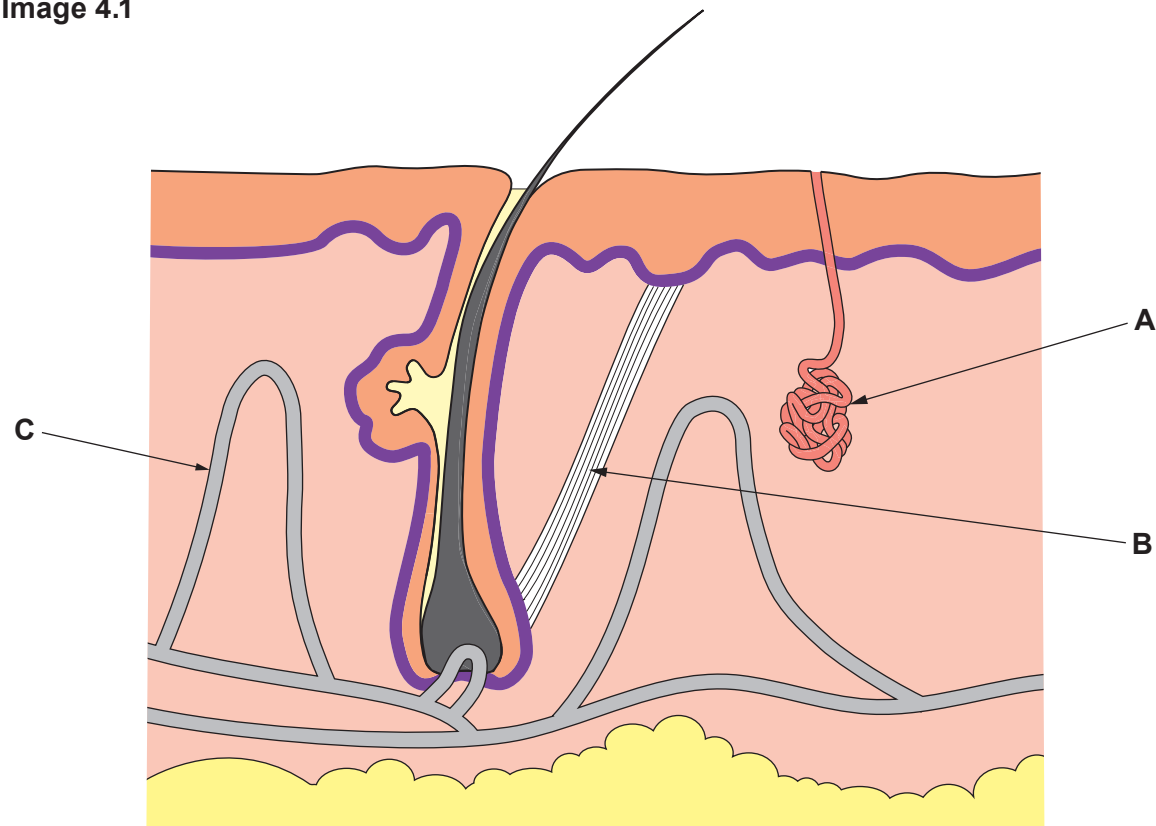
- (a) (i) Using the five-kingdom system of classification, name the kingdom to which *Staphylococcus aureus* belongs. [1]
-
- (ii) State what is meant by the term methicillin resistant *Staphylococcus aureus*. [1]
-
-
- (b) (i) Describe the trend in the numbers of deaths due to MRSA in Wales and England between 1993 and 2012. [1]
-
-
- (ii) Suggest **one** explanation for the trend in the numbers of deaths due to MRSA in Wales and England between 2006 and 2012. [1]
-
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07



4. Image 4.1 shows a section through the human skin.

Image 4.1



(a) State the name of structures **A** and **B** in Image 4.1. [1]

A

B

(b) Explain how structure **B** helps to maintain the normal body temperature in cold conditions. [3]

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Examiner
only

- (c) Frostbite occurs when the skin and tissues are damaged by long-term exposure to freezing temperatures. The most common parts of the body that frostbite affects are the fingers, toes, nose, ears, cheeks and chin.
Using **Image 4.1** suggest how the response of the blood vessels labelled **C** to freezing temperatures can lead to frostbite developing in the fingers. [2]

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Examiner
only

5. Scientists conducted DNA profiling of the critically endangered smalltooth sawfish (*Pristis pectinata*).



The results of the DNA profiling showed that a number of female sawfish had been able to reproduce without mating. All of the offspring from these fish were female.

- (a) Describe the process of DNA profiling. [2]

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- (b) State the name of the type of reproduction carried out by the female sawfish described above and explain why all of their offspring were female. [2]

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4



6. The GloFish is a genetically modified Zebrafish which has become popular in aquariums in the United States and parts of Asia.

The fish were produced by inserting a gene from a jellyfish into a Zebrafish. The gene controlled the production of a fluorescent protein (GFP). The fish produced were brightly coloured.

GloFish are not approved for sale in Australia, Canada, California or Europe. **Image 6.1A** shows a Zebrafish and **Image 6.1B** shows a GloFish.

Image 6.1A



Image 6.1B



- (a) (i) State the name of the molecule that makes up genes. [1]

- (ii) Explain how the gene controls the production of the GFP protein. [3]

- (b) Suggest why GloFish are not approved for sale in some parts of the world. [1]

3400UD01
11



7. According to the World Health Organisation, the failure of parents to vaccinate their children in 2016 led to the biggest surge in measles cases in Europe for over a decade. Across 53 countries, there were 37 deaths reported from a total of 41 000 cases of measles.

(a) (i) Calculate the death rate for measles per 1000 cases of measles. [1]

Death rate = per 1000 cases of measles

(ii) Suggest a reason why some parents choose not to have their children vaccinated despite the risk to their health. [1]

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(b) Explain why an individual who has contracted measles once is unlikely to contract the disease a second time. [4]

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(c) Typhoid is a serious disease caused by the bacterium *Salmonella typhi*. Individuals who travel to the Indian subcontinent, Africa, South America and southeast Asia are advised to be vaccinated. The typhoid vaccine contains fragments of *S. typhi*.

(i) Explain why an individual receiving the typhoid vaccine will not develop the disease even though it contains fragments of *S. typhi*. [1]

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Examiner
only

(ii) Explain why the typhoid vaccine does not prevent fevers or infections caused by other species of *Salmonella* bacteria. [1]

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(d) State how an increased understanding of the human genome can be important for medicine. [1]

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9

8. Diabetes is a condition where the concentration of glucose in the blood can become too high. Over time, this may cause complications including damage to blood vessels and nerves.

(a) (i) Apart from high blood glucose levels, state **one** other symptom of diabetes. [1]

(ii) Insulin is essential in the control of blood glucose. With reference to insulin, explain the difference between Type 1 and Type 2 diabetes. [2]

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(iii) Apart from insulin injections, describe **one** other possible treatment for people with diabetes. [1]

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(b) Explain how a negative feedback mechanism raises blood glucose levels in a healthy individual when their glucose levels are decreasing. [3]

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- (c) **Table 8.1** and **Table 8.2** show information on the ethnic categories of 26 907 children and young people up to the age of 24 years in Wales and England in 2016/17 who were diagnosed with either Type 1 or Type 2 diabetes.

Table 8.3 shows the ethnic categories of **all** children and young people in Wales and England.

Table 8.1 – Ethnic categories of children and young people with **Type 1 diabetes, 2016/17**

Ethnic category	Number	Percentage of the sample of children who had Type 1 diabetes (%)
White	22 111	84.0
Mixed	699	2.7
Asian	2 136	8.1
Black	956	3.6
Other	421	1.6
Total number of children with Type 1 diabetes =	26 323	

Table 8.2 – Ethnic categories of children and young people with **Type 2 diabetes, 2016/17**

Ethnic category	Number	Percentage of the sample of children who had Type 2 diabetes (%)
White	246	42.1
Mixed	29	5.0
Asian	212	36.3
Black	79	13.5
Other	18	3.1
Total number of children with Type 2 diabetes =	584	



Examiner
only

Table 8.3 – Ethnic categories of **all children and young people in Wales and England**

Ethnic category	Percentage of population of all children and young people in Wales and England (%)
White	86.0
Mixed	2.2
Asian	7.5
Black	3.3
Other	1.0

- (i) Calculate the ratio of the total number of children and young people with Type 1 diabetes against the total number with Type 2 diabetes. [1]

Type 1 : Type 2

- (ii) Use **Table 8.1**, **Table 8.2** and **Table 8.3** to state **one** conclusion that can be drawn on the effect of ethnicity on the percentage of children and young people affected by:

I. Type 1 diabetes; [1]

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II. Type 2 diabetes. [1]

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- (d) A politician looking at these data concluded that the number of people in the whole population of Wales and England with Type 1 diabetes is far higher than the number with Type 2 diabetes. State why this is not a valid conclusion. [1]

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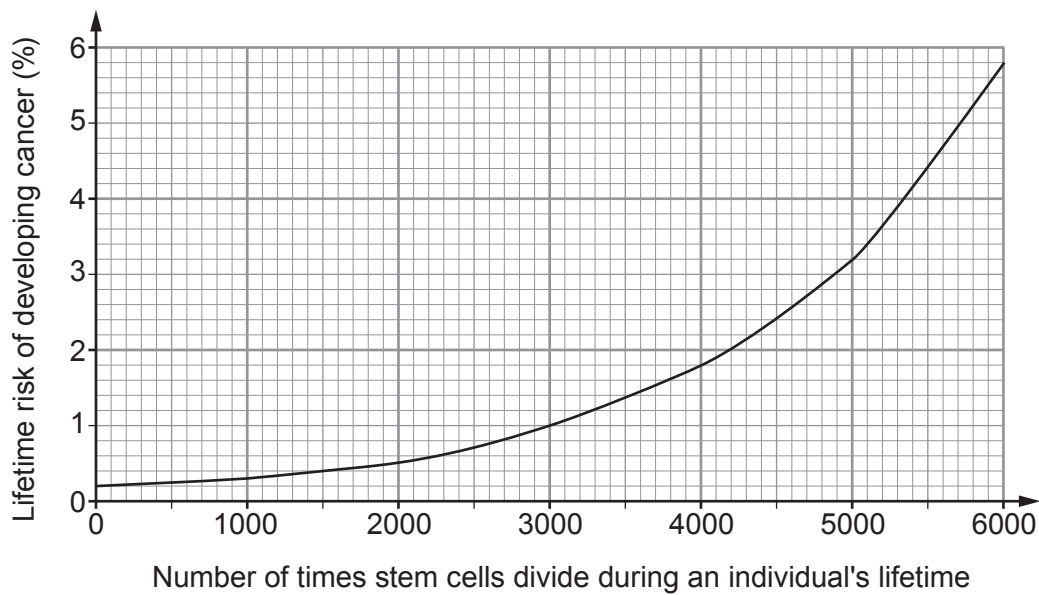
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9. In humans, the number of times a stem cell undergoes cell division varies. For example, some stem cells in the skin will divide up to 200 times during an individual's lifetime whilst other stem cells found in the large intestine may divide up to 5800 times. The lifetime risk of developing a cancer is different for each type of tissue in the human body. For example, the lifetime risk of developing skin cancer is 2%. The lifetime risk of developing a cancer of the liver is 0.71%. Scientists have suggested that some of the variation in cancer risk can be explained by the number of times stem cells divide in each tissue during an individual's lifetime, as shown in **Graph 9.1**.

Graph 9.1



- (a) (i) State what is meant by a stem cell. [2]

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- (ii) State the name of the type of cell division responsible for growth and explain how it can lead to cancer. [2]

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Examiner
only

- (b) (i) Use all the information given to identify the percentage lifetime risk of developing cancer of the large intestine. [1]

Lifetime risk = %

- (ii) Suggest **one** reason why the incidence of cancer is rare amongst young people, but tends to increase with age. [1]

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- (c) Other than environmental factors and the number of times a cell divides, name **one** other factor which could be responsible for the variation in the risk of cancers in different tissues. [1]

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7



10. Urine samples from four individuals were tested in a laboratory. Some of the results are shown in **Table 10.1**. ✓ = present, ✕ = absent.

Table 10.1

Sample	Sodium chloride (salt)	Glucose	Protein
1	✓	✕	✕
2	✓	✓	✕
3	✓	✕	✓
4	✓	✓	✓

(a) (i) State the name of **one** other substance that you would expect to be present in each of the samples. [1]

(ii) Explain why you would not expect to find either protein or glucose in the urine of a healthy individual. [2]

(b) State the name of the solution you would use to test for protein in a school laboratory and state the expected colour change if sample 3 in **Table 10.1** was tested for protein. [2]



- (c) Kidney failure can be treated using dialysis. **Image 10.2** shows a patient connected to a simplified drawing of a dialysis machine.

Image 10.2

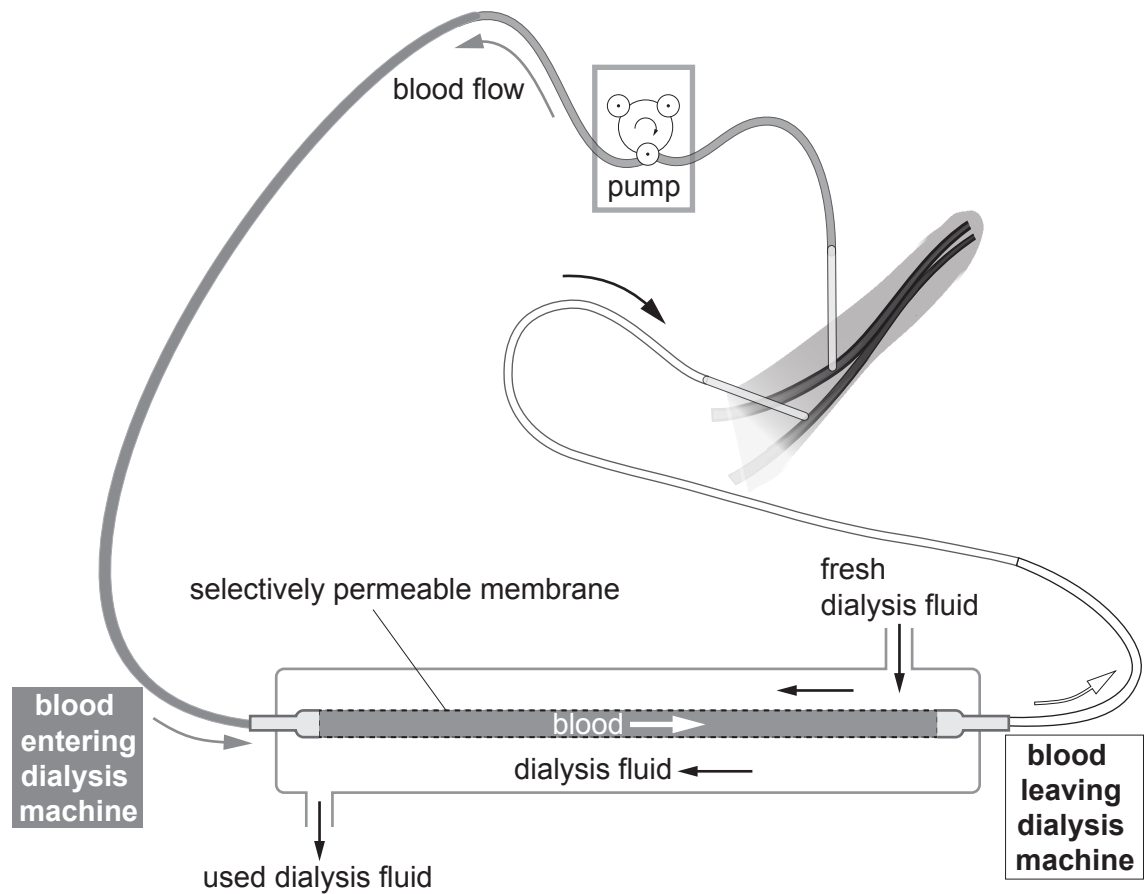


Table 10.3 compares the composition of blood and dialysis fluid.

Table 10.3

Substance	Concentration (arbitrary units)		
	Blood entering dialysis machine	Fresh dialysis fluid	Used dialysis fluid
Sodium ions	140	130	150
Chloride ions	100	90	110
Glucose	100	100	100
Urea	20	0	18
Protein	4	0	0



Examiner
only

- (i) The pump in the dialysis machine pumps the blood at a rate of 400 cm^3 **per minute**. Calculate the volume of blood in dm^3 that would pass through the dialysis machine if a patient was connected to the machine for **four hours**. ($1000 \text{ cm}^3 = 1 \text{ dm}^3$)

Volume of blood = dm³

- (ii) Explain why the blood and the dialysis fluid flow in opposite directions through the machine. [1]

- (iii) Using all the information given, explain how the process of dialysis takes place in the kidney machine. [6 QER]

END OF PAPER

14



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE

3400UB0-1



TUESDAY, 17 MAY 2022 – MORNING

BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

HIGHER TIER
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	9	
2.	11	
3.	8	
4.	9	
5.	7	
6.	9	
7.	9	
8.	11	
9.	7	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 7(b) is a quality of extended response (QER) question where your writing skills will be assessed.



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Answer **all** questions.

1. Biological control is used to reduce the numbers of a pest population through the introduction of another species. It has been used with varying success since the 19th century.

(a) (i) State **two** advantages of this method of control. [2]

Advantage 1

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.....

Advantage 2

.....

.....

(ii) State **two** disadvantages of this method of control. [2]

Disadvantage 1

.....

.....

Disadvantage 2

.....

.....

(b) The whitefly (*Trialeurodes vaporariorum*) is a pest which damages greenhouse crops such as tomatoes. Whitefly numbers can be reduced by using the biological control agent *Encarsia formosa*.

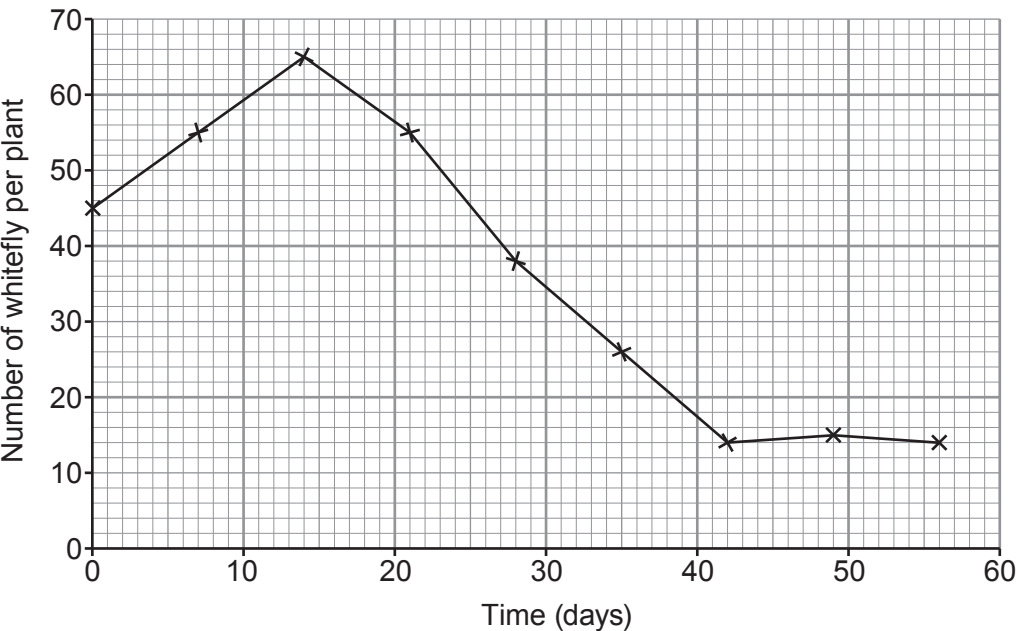
***Encarsia formosa* fact file**

- *E. formosa* is a tiny wasp that lays eggs inside developing whitefly.
- When the eggs hatch, the young wasps kill the developing whitefly from the inside.
- Optimal conditions for *E. formosa* are temperatures over 20°C.
- When daytime temperatures are less than 17°C, *E. formosa* activity is significantly reduced, making it less effective.



Graph 1.1 shows the number of whiteflies in a greenhouse containing tomato plants. *E. formosa* were introduced on day 7.

Graph 1.1



- (i) I. The use of *E. formosa* to reduce the number of whiteflies is considered to be successful when there are 20 or fewer whiteflies per plant. Use **Graph 1.1** to determine how many days it took for the number of whiteflies to fall to 20 following the introduction of *E. formosa*. [2]

..... days

- II. Suggest a reason why it took this long for the number to fall to 20. [1]

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- (ii) Suggest **one** reason why this method of pest control would not be effective to use if whiteflies damaged wheat crops grown in Wales. [1]

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- (iii) An alternative approach to reducing pest numbers is to use pesticide. State why it is not appropriate to use pesticide along with *E. formosa*. [1]

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2. Hair length in cats is controlled by a pair of alleles. The allele for short hair (**H**) is dominant to the allele for long hair (**h**).



(a) State what is meant by the terms:

(i) allele; [1]

.....

.....

(ii) dominant; [1]

.....

.....

(iii) recessive. [1]

.....

.....



- (b) (i) A cat breeder crossed a homozygous short-haired cat with a long-haired cat. **Complete the Punnett square** to show the predicted genotypes of the offspring. Use the letters **H** and **h** for the alleles. [2]

Gametes		

- (ii) State the **phenotype** of the offspring in the F1 generation. [1]

.....

- (iii) **Complete the Punnett square** to show the possible genotypes of the offspring if two of the F1 offspring were crossed. [2]

Gametes		

- (iv) Using the results from (b)(iii), state how many kittens would be predicted to be short-haired in a litter of 8 kittens. [1]

.....

- (v) The cat breeder wanted to determine whether one of the short-haired cats was homozygous or heterozygous. She decided to breed the short-haired cat with a long-haired cat. Predict the phenotypes of the offspring you would expect if the short-haired cat was:
- I. Homozygous [1]

.....

- II. Heterozygous [1]

.....



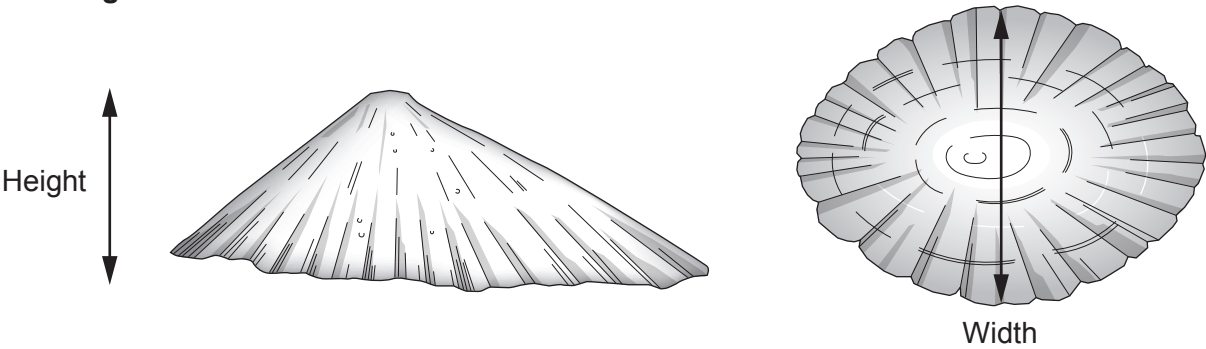
3. The common limpet (*Patella vulgata*) is found attached to rocks on most rocky shores in Wales. Limpets attach firmly to the rocks using a muscular foot. They also rotate their shell and grind it into rock to seal it into the rock. Limpets vary in height and width.



The common limpet

A survey was carried out to investigate the variation in the size of limpet shells on upper and lower shores in Cemlyn Bay, Anglesey in North Wales. The height and width of each shell were measured to the nearest mm as shown in **Image 3.1**.

Image 3.1



These measurements were used to calculate the height : width ratio.

$$\text{Ratio} = \frac{\text{height}}{\text{width}} : 1$$

- (a) (i) State what is meant by the term variation. [2]

.....

.....

- (ii) State the type of variation shown in the size of the limpets. [1]

.....



Table 3.2

Limpet number	Limpet shell height : width ratio	
	Upper shore	Lower shore
1	1.35 : 1	0.38 : 1
2	1.53 : 1	0.36 : 1
3	1.47 : 1	0.41 : 1
4	1.80 : 1	0.28 : 1
5	3.11 : 1	0.44 : 1
6	3.00 : 1	0.57 : 1
7	2.56 : 1	0.58 : 1
8	2.45 : 1	0.37 : 1
9	2.00 : 1	0.27 : 1
10	2.42 : 1	0.30 : 1
Mean	2.17 : 1	 : 1

(b) (i) Calculate the mean shell height : width ratio for limpets sampled on the lower shore. **Write your answer in Table 3.2.** [2]
Space for working

(ii) State the conclusion that can be made about the height to width ratios of limpets and their position on the shore. Suggest a reason for this. [2]

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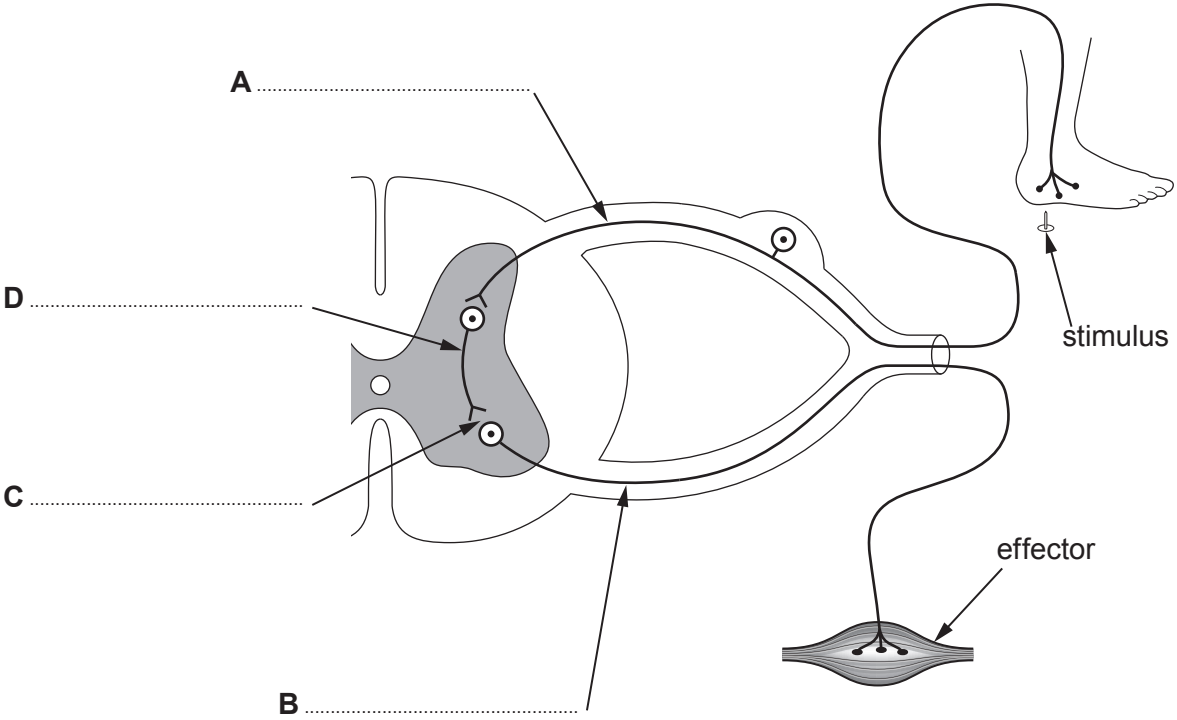
(iii) State how the students could increase confidence in their results. [1]

.....

.....



4. Image 4.1 shows a reflex arc.



(a) (i) Label **A–D** on **Image 4.1**.

[4]

(ii) State **two** properties of reflex actions.

[1]

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.....



Examiner
only

- (iii) The length of neurone **B** is 0.9 m. An electrical impulse can travel along a neurone at 75 m/s.

Use the following equation:

$$\text{Speed} = \frac{\text{distance}}{\text{time}}$$

to calculate the time taken for the impulse to travel the length of neurone **B**. [2]

Time =s

- (b) Motor neurone disease prevents motor neurones functioning effectively.
Explain why individuals with the disease find it difficult to walk. [2]

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9

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5. The yellow crazy ant (*Anoplolepis gracilipes*) was introduced accidentally to northern Australia. Due to the ecological damage it has caused, it is on a list of “one hundred of the world’s worst invasive species” produced by the International Union for Conservation of Nature (IUCN). The ants build super-colonies that disrupt native habitats.



Yellow crazy ants

- (a) (i) State what is meant by the term invasive species. [1]

.....

.....

- (ii) State the genus of the yellow crazy ant. [1]

.....

- (iii) Explain how the formation of super-colonies affects biodiversity. [1]

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Examiner
only

(b) Female ants have a chromosome number of 34 and produce gametes through the process of meiosis.

(i) State how many gametes are produced from each mother cell. [1]

.....

(ii) Males develop from female eggs which have not been fertilised. Conclude the chromosome number in males and explain your answer. [2]

.....

.....

(iii) State **one** function of mitosis in adult ants. [1]

.....

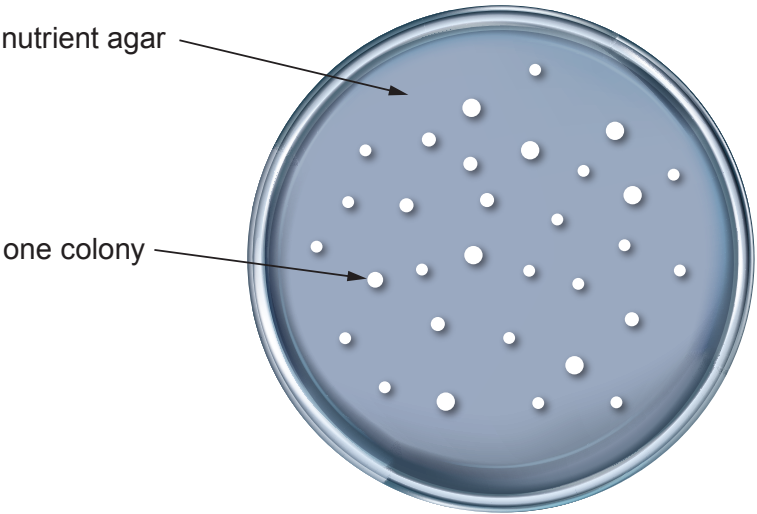
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6. Environmental health officers estimated the number of bacteria in a milk sample from a cafe. The limit for the total number of bacteria in a sample of milk which is considered to be safe for human consumption is 1.0×10^5 bacteria per cm^3 .
- Using aseptic techniques, they diluted the sample by a factor of 10 000.
 - They plated 1 cm^3 of the diluted sample onto nutrient agar.
 - The lid of the plate was secured with tape.
 - The plate was incubated at 37°C for 2 days.
 - The result is shown in **Image 6.1**.

Image 6.1



- (a) (i) State the assumption that must be made when calculating the number of bacterial cells present in the original sample. [1]

.....

.....

- (ii) I. Calculate the number of bacteria in 1 cm^3 of the original sample taken by the environmental health officers. **Write your answer in standard form.** [3]

Number of bacteria =

- II. Conclude whether the milk sample was safe for human consumption. Explain your answer. [1]

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Examiner
only

(b) Suggest why the plates were incubated at 37 °C. [1]

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.....

(c) (i) State why the lids of the agar plates were secured with tape. [1]

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(ii) State **two** other precautions that should have been taken to ensure aseptic technique. [2]

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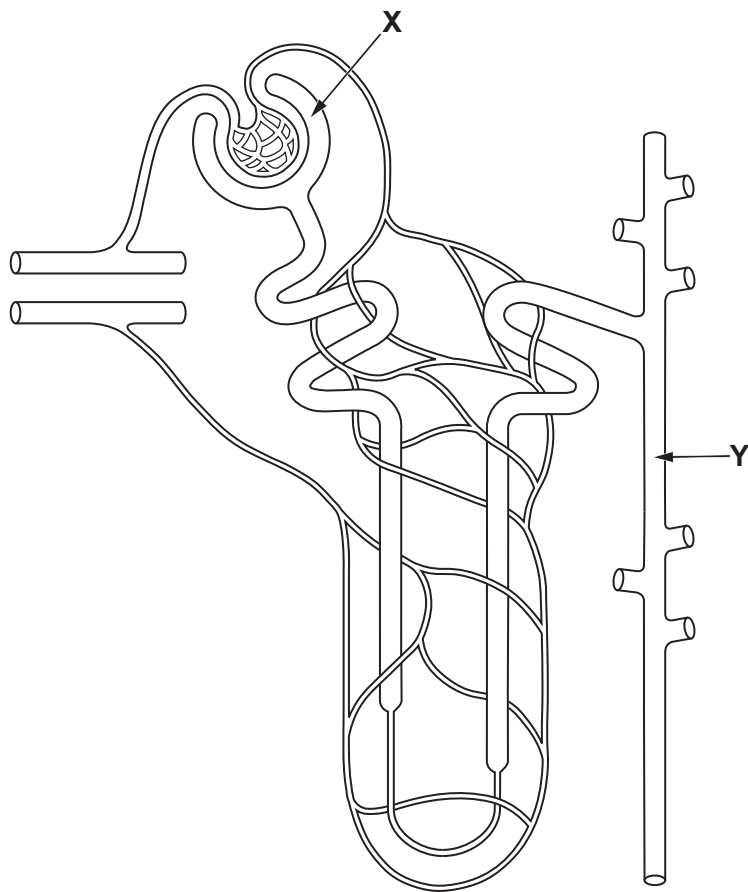
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7. Image 7.1 shows the structure of a nephron.

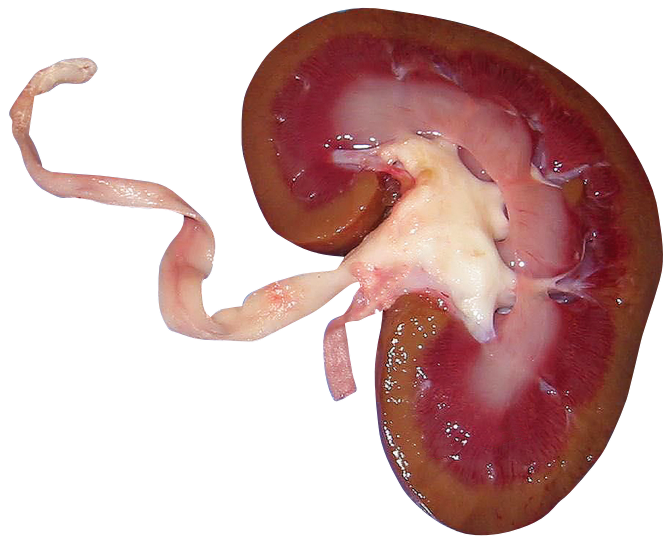
Image 7.1



- (a) Use a labelled line to show the position of structure X (shown in Image 7.1) in the section through the kidney in Image 7.2.

[1]

Image 7.2



Examiner
only

- (b) The filtrate found at **X** in **Image 7.1** differs from the filtrate found at **Y** in **Image 7.1**. Describe and explain the composition of the filtrate at **X** and **Y**, identifying structures **X** and **Y** in your answer.
(No reference to ADH is required) [6 QER]

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- (c) Protein is found in the blood in the capillary knot but is not found in the filtrate at **X**. Explain this observation. [2]

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9



8. There are many different types of cancer. One of the most common is breast cancer. In approximately 30% of breast cancers a high level of HER2 protein is present, this is due to a mutation in the HER2 gene. The high level of HER2 protein found on the surface of the cells causes them to grow and divide excessively.

(a) State what is meant by the term mutation. [1]

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.....

(b) (i) In Wales, approximately 2900 people are diagnosed with breast cancer each year. Calculate how many of those diagnosed would be expected to have a high level of HER2 protein present. [2]

Number of people =

(ii) Explain how the HER2 gene codes for the production of HER2 protein. [3]

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Examiner
only

- (c) Since 1985, approximately 100 different monoclonal antibodies have been developed and some are used in the treatment of breast cancer linked to the mutated HER2 gene. During chemotherapy a drug is attached to the monoclonal antibody to target these cancer cells directly.

Explain how the monoclonal antibody targets these cancer cells directly. [3]

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- (d) State **two** other uses of monoclonal antibodies. [2]

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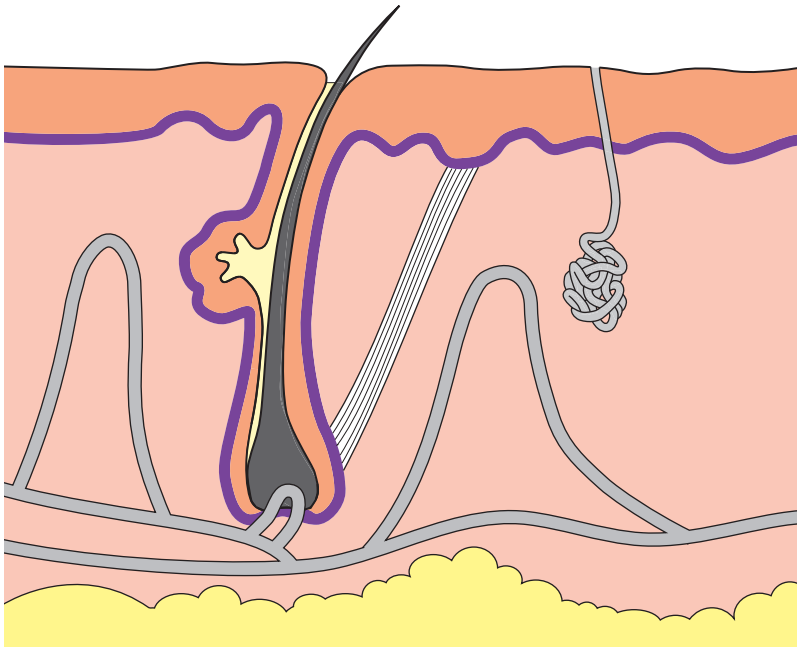
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11



9. Image 9.1 shows a section through human skin.

Image 9.1



(a) On Image 9.1, draw labelled arrows to name and identify:

(i) the structure where sweat is produced;

[1]

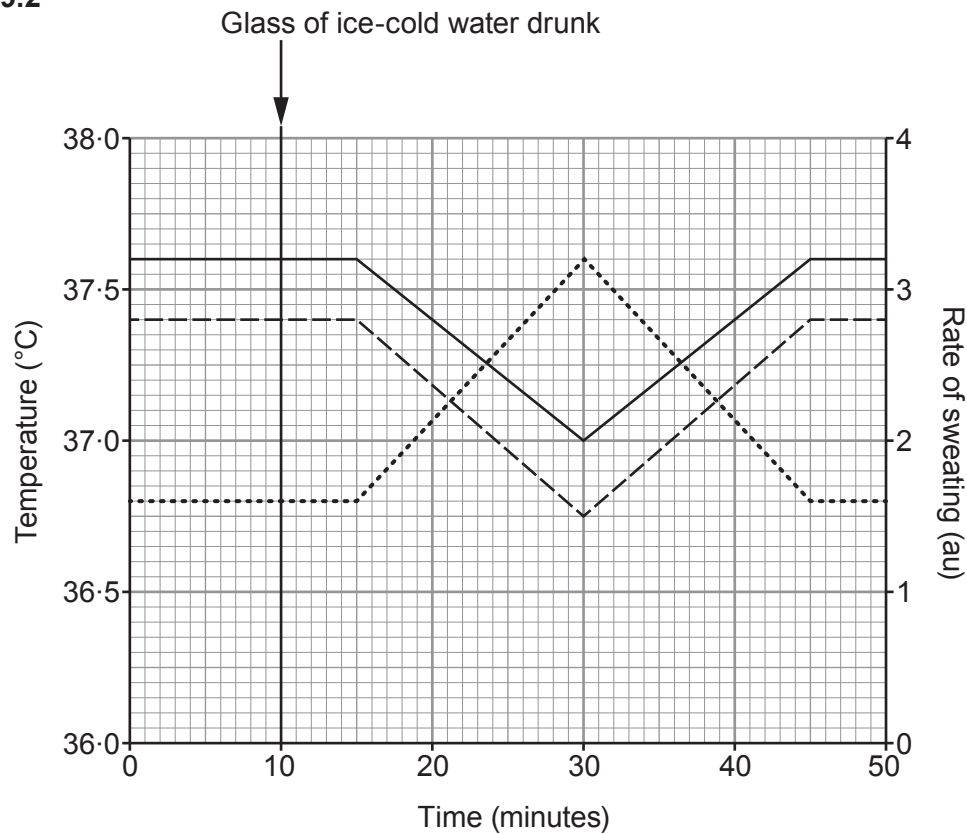
(ii) a structure which transports blood.

[1]



Gareth carried out an experiment to investigate homeostasis. He sat in a room heated to 50°C. His internal body temperature, rate of sweating and skin temperature were monitored. After 10 minutes in the room, he drank a glass of ice-cold water. The results are shown in **Graph 9.2**.

Graph 9.2



- Key:**
- Internal body temperature
 - Rate of sweating
 - Skin temperature



Examiner
only

(b) Explain the effect that drinking the ice-cold water had on Gareth between 10 minutes and 30 minutes, as shown on **Graph 9.2**. [5]

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END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3400UB0-1



TUESDAY, 16 MAY 2023 – MORNING

BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms
HIGHER TIER
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	9	
3.	4	
4.	5	
5.	6	
6.	8	
7.	8	
8.	17	
9.	12	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **9(b)** is a quality of extended response (QER) question where your writing skills will be assessed.

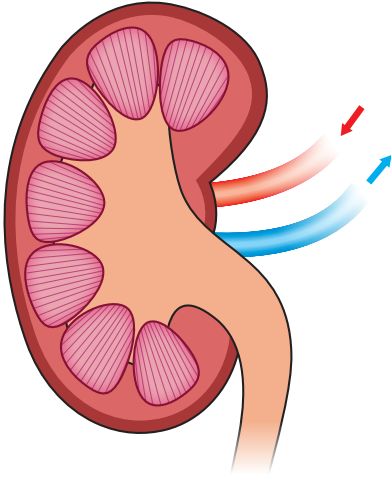


JUN233400UB0101

Answer **all** questions.

1. **Image 1.1** shows a section through a human kidney.

Image 1.1



- (a) **Draw arrows on Image 1.1** to label the following parts: [2]
- (i) the medulla;
 - (ii) the renal artery.

(b) **Table 1.2** shows the results of an investigation on a patient in a hospital.

Table 1.2

Substance	Concentration in blood entering kidney (au)	Concentration in blood leaving kidney (au)	Present in urine (✓ or ×)
glucose	145	32	✓
protein	120	120	
urea	93	0	

- (i) I. **Complete Table 1.2** to show which substance(s) would be present in the urine of this patient. One row has been done for you. [1]
- II. State which **one** of the substances is a waste product. [1]
-
- III. Doctors suggested that this patient had diabetes. From **Table 1.2**, state the evidence to support this suggestion. [1]
-
-



Examiner
only

(ii) Describe the chemical tests which could be carried out to confirm whether or not the urine contained glucose and protein. For each test state the colours you would expect **for the results shown in Table 1.2.**

I. glucose test [3]

.....

.....

.....

II. protein test [2]

.....

.....

(iii) **Complete the risk assessment** below which shows one hazard linked to these tests. [1]

Hazard	Risk	Control measure
Chemical reagent is an irritant		
		
		
		

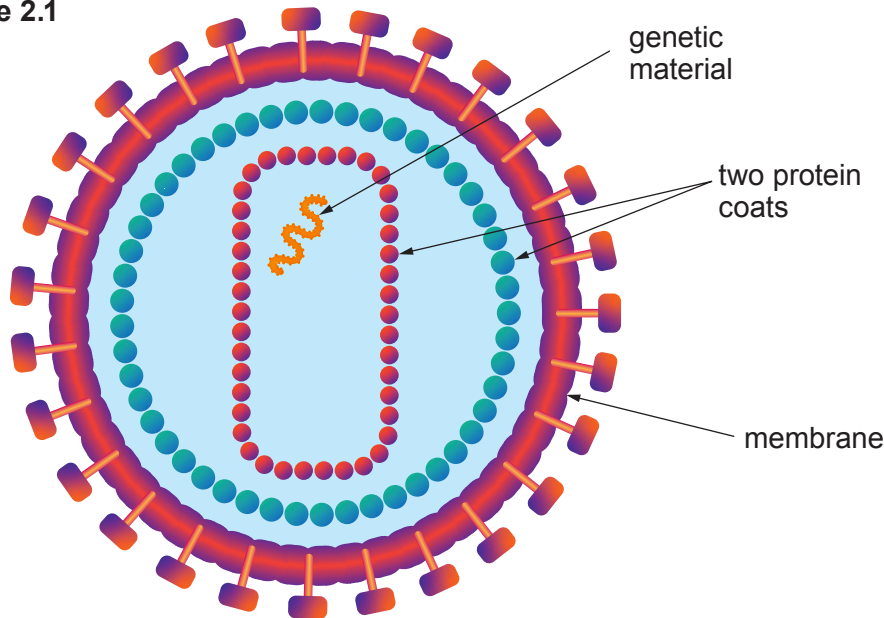
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2. **Image 2.1** shows the structure of the human immuno-deficiency virus (HIV).

In some of the people who have an HIV infection, the virus can lead to the disease AIDS, which can be fatal.

Image 2.1



(a) State **two** ways, visible in **Image 2.1**, in which the structure of HIV is different from other viruses. [1]

1.
2.

(b) State how HIV is spread from one person to another and **one** measure which can be taken to reduce the spread. [2]

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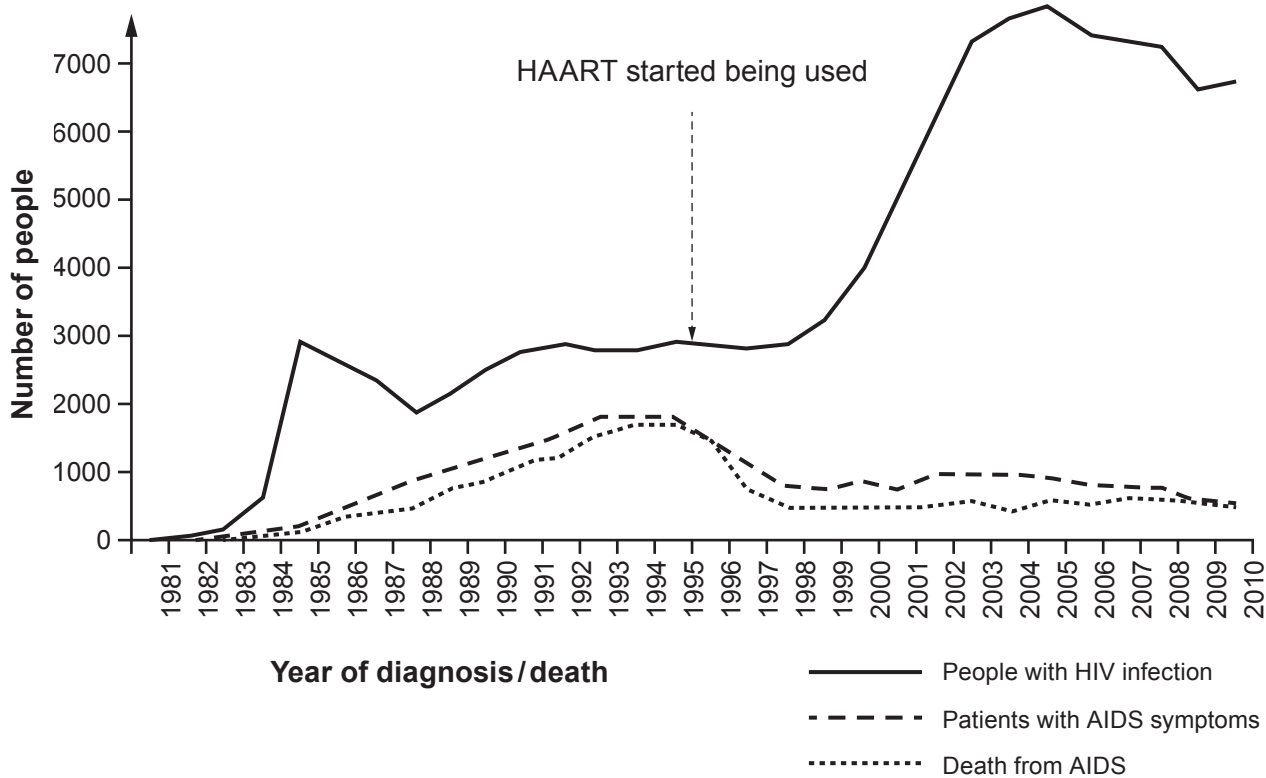


- (c) In 2000, the United Nations set a target that 90% of people with HIV should be receiving treatment. This treatment should be effective at reducing the number of people developing AIDS.

By 2010, 97% of people with HIV in the UK were being treated with the anti-viral drug, HAART.

Graph 2.2 shows the results from investigations on the effectiveness of HAART.

Graph 2.2



Using the information provided on this page, give the evidence that by 2010 the UK had met the UN target by

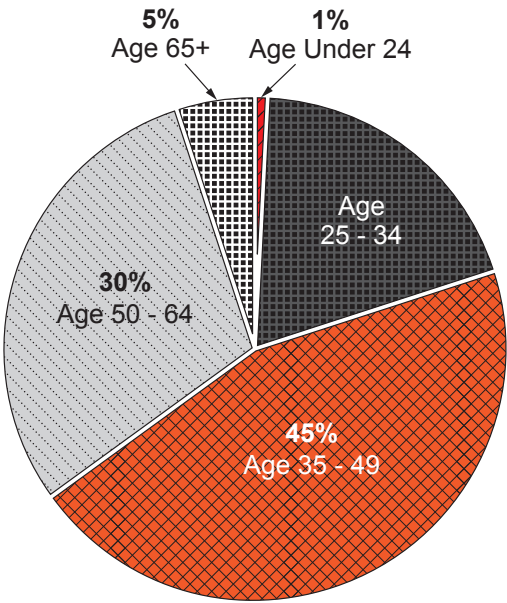
- (i) providing treatment to enough people; [1]

- (ii) providing treatment which was effective. [1]



- (d) **Graph 2.3** shows the percentages of people in different age groups in the UK who were known to be infected with HIV in 2017. The total number of infected people was 101 610.

Graph 2.3



People infected with HIV in the UK

From the information given above calculate

- (i) **the percentage** of people in the 25–34 age group, [1]

..... %

- (ii) **the number** of people in the 35–49 age group. [2]

..... people.

- (iii) Most campaigns which encourage people to take treatment for HIV infections are targeted at those under 35. Some charities, however, say it would be more effective to target campaigns at older age groups.

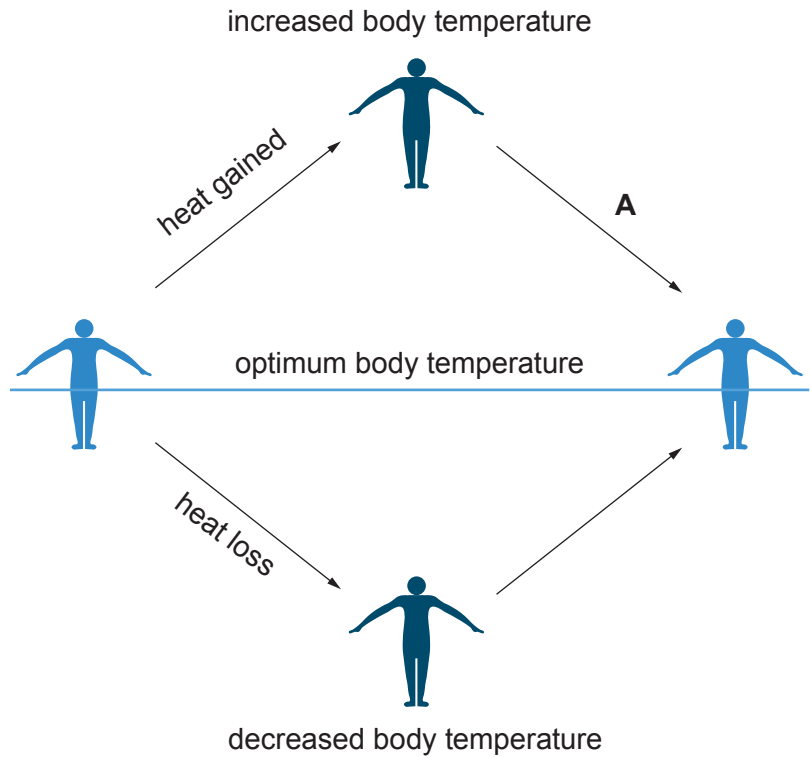
Using the information in **Graph 2.3**, state **one** piece of evidence which supports the point of view of some charities. [1]

.....



3. (a) **Image 3** shows how the optimum body temperature is maintained.

Image 3



(i) State the name of the **mechanism** which maintains optimum body temperature.[1]

.....

(ii) State why it is important that human body temperature remains between 36.5°C and 37.5°C. [1]

.....
.....

(iii) Apart from sweating, describe and explain **one** way the skin would have responded at point **A** on **Image 3** to return the body temperature to its optimum. [2]

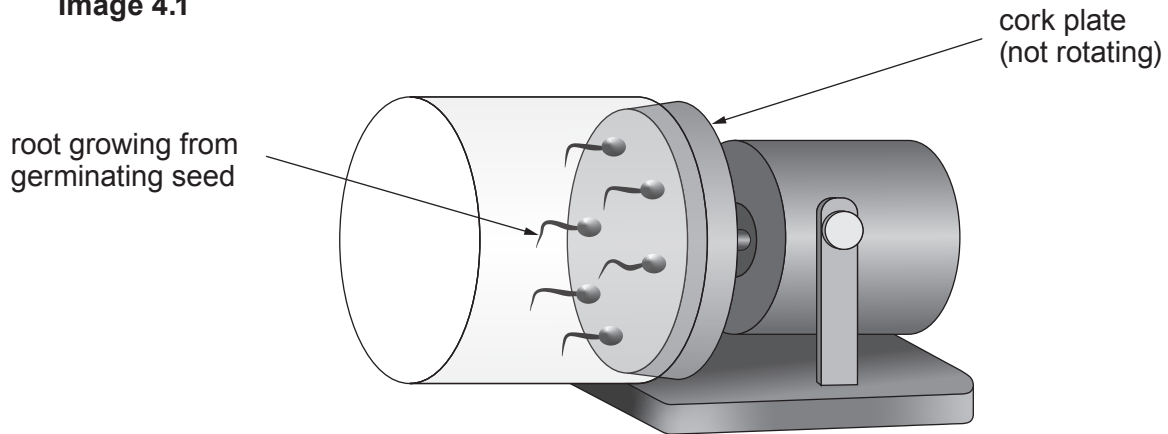
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4. In a school experiment, germinating seeds were pinned to a cork plate (**which was not rotating**) and left in the dark for 48 hours.

The result of the experiment is shown in **Image 4.1**.

Image 4.1



- (a) State the name of the:

- (i) growth response shown by the germinating roots in **Image 4.1**;

[1]

.....

- (ii) hormone which controls growth responses in plants.

[1]

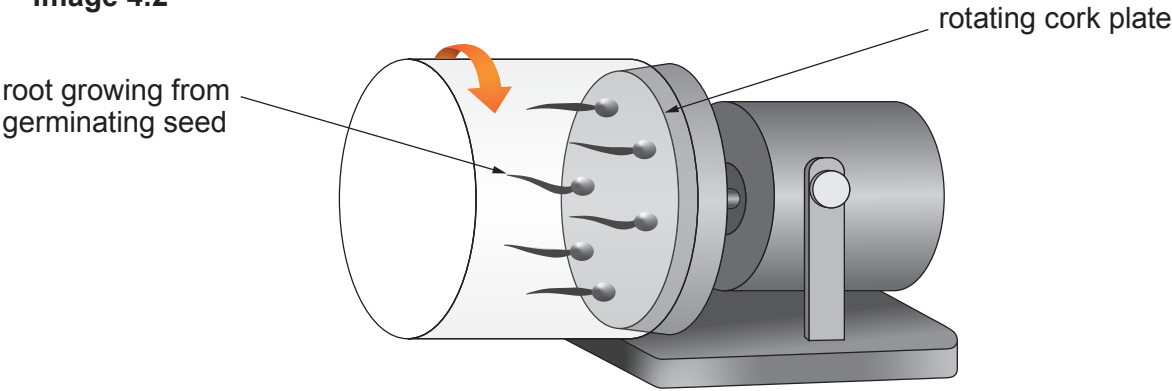
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A second experiment was set up where germinating seeds were pinned to a cork plate which rotated three times per hour. The apparatus was left in the dark for 48 hours.

The result of the experiment is shown in **Image 4.2**.

Image 4.2



(b) Explain the growth of the roots in the result shown in **Image 4.2**. [2]

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(c) Suggest why the apparatus was kept in the dark for 48 hours. [1]

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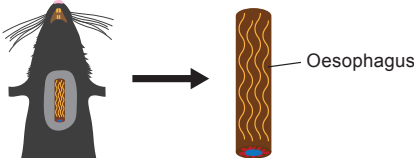
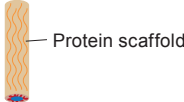
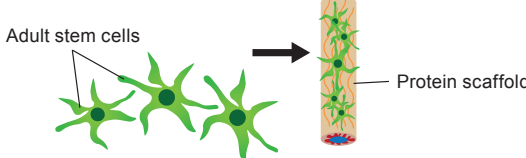
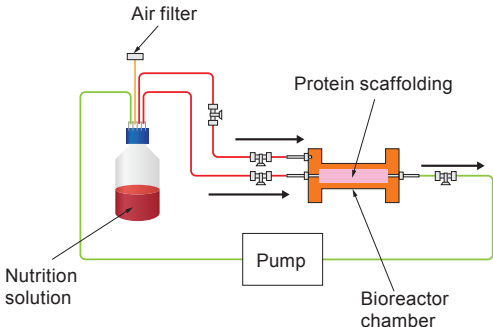
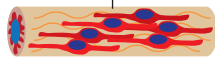
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5. In 2018, scientists announced they had produced a laboratory grown oesophagus using a rat oesophagus and human adult stem cells. This was part of on-going research by scientists into different ways of producing organs for transplants.

Image 5 outlines the procedure.

Image 5

Step 1: An oesophagus was removed from a rat	 The diagram shows a rat on the left with a yellow oesophagus highlighted inside its body. An arrow points to the right, where a separate, elongated yellow oesophagus is shown. A label 'Oesophagus' points to this structure.
Step 2: All the rat cells were removed from the oesophagus leaving only a protein scaffold	 The diagram shows a single, elongated yellow protein scaffold. A label 'Protein scaffold' points to it.
Step 3: Human adult stem cells were added to the protein scaffold	 The diagram shows several green, star-shaped 'Adult stem cells' on the left. An arrow points to the right, where these cells are shown attached to the yellow 'Protein scaffold'. A label 'Protein scaffold' points to the scaffold.
Step 4: The protein scaffold with human adult stem cells was placed in a bioreactor. A bioreactor is a machine which maintains optimum conditions for cell growth	 The diagram shows a bioreactor system. On the left, a bottle of 'Nutrition solution' is connected by a red tube to a 'Pump'. The pump feeds the solution into a 'Bioreactor chamber' which contains the 'Protein scaffolding'. An 'Air filter' is also connected to the top of the chamber. Arrows indicate the flow of the nutrition solution and air into the chamber.
Step 5: At 11 days, the oesophagus was removed from the bioreactor	 The diagram shows the final 'Laboratory grown oesophagus', which is a yellow tube with red internal structures. A label 'Laboratory grown oesophagus' points to it.



Examiner
only

(a) (i) State the meaning of the term stem cell. [1]

.....

.....

(ii) State the name of the type of cell division which occurs during **Step 4**. [1]

.....

(iii) Suggest **one** advantage of using adult stem cells rather than embryonic stem cells in the process shown in **Image 5**. [1]

.....

.....

(iv) Suggest **two** variables which need to be controlled in the bioreactor. [2]

.....

.....

(b) State **one** reason why animal-rights groups might oppose this type of research. [1]

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6



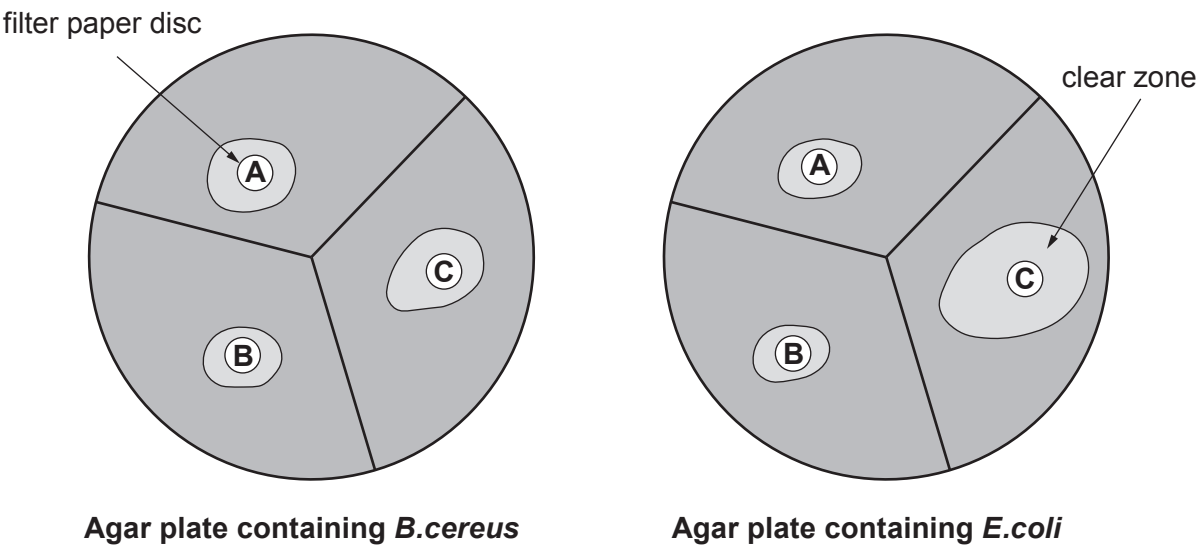
6. A class of students compared the effects of two antibiotics and an extract from the seeds of moringa plants on the growth of bacteria. Each group of students was given two agar plates. One plate had bacterium *B.cereus* growing on it whilst the other had the bacterium *E.coli* growing on it. The students used aseptic techniques to place three filter paper discs labelled **A**, **B** and **C** on each plate. Each disc had each been soaked in one of the following solutions as shown in **Table 6.1**.

Table 6.1

Filter paper disc	Solution
A	moringa seed extract
B	penicillin (an antibiotic)
C	tetracycline (an antibiotic)

The plates were incubated for 24 hours. The results are shown in **Image 6.2**.

Image 6.2



The students measured the diameter of the clear zones and recorded them in **Table 6.3**.

Table 6.3

Bacterium	Diameter of clear zone (mm)		
	Disc A moringa seed extract	Disc B penicillin	Disc C tetracycline
<i>B.cereus</i>	11	8	12
<i>E.coli</i>	9	8	18



Examiner
only

(a) (i) State **two** steps the students would have used to maintain aseptic conditions whilst placing the discs on each plate. [2]

.....

.....

.....

.....

(ii) State a suitable temperature the students would have used to incubate the agar plates in a school laboratory. [1]

..... °C

(b) (i) Explain how the clear zones formed on the agar plates shown in **Image 6.2**. [2]

.....

.....

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(ii) State **two** conclusions which can be made about the effect of the moringa seed extract on the growth of bacteria compared to the two antibiotics. [2]

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(c) Suggest **one** way in which each group of students in the class could have increased the validity of their conclusions. [1]

.....

.....

8



7. Barnacles are small animals that grow on rocks at the seashore. Two species of barnacle native to the British Isles, *Chthamalus stellatus* and *Semibalanus balanoides* are shown in the photographs in **Image 7.1**.

Image 7.1



C. stellatus



S. balanoides

Students carried out a survey to test the following hypothesis:

- C. stellatus* is found further up the seashore towards the high water mark.
S. balanoides is found lower down the seashore towards the low water mark.

Image 7.2



Photograph of students carrying out their survey



The students measured the abundance and distribution of barnacles between the high water mark and the low water mark on the seashore. The results of their survey are shown in **Table 7.3**.

Table 7.3

Distance from high water mark (m)	Number of barnacles	
	<i>C. stellatus</i>	<i>S. balanoides</i>
0 (high water mark)	0	0
5	13	0
10	28	0
15	36	0
20	52	3
25	57	25
30	29	41
35	17	51
40	0	60
45	0	42
50 (low water mark)	0	0

- (a) Describe the method the students would have used to measure the abundance and distribution of barnacles between the high and low water mark. [3]

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Examiner
only

(b) (i) State **two environmental** factors which could affect the population of barnacles. [2]

I.

II.

(ii) Calculate the ratio of *C. stellatus* to *S. balanoides* at the **25 m** mark. [1]
Space for working.

C. stellatus : *S. balanoides*

(iii) State the type of competition occurring between the two species of barnacles at the 25 m mark. [1]

.....

(iv) State whether the results of the survey support the students' hypothesis (given on page 15). Use the data provided in **Table 7.3** to give a reason for your answer. [1]

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8



8. Haemochromatosis is a genetic condition which causes the body to absorb more iron than normal from the diet. The excess iron is stored in the body's tissues and organs where it can cause damage. The condition mainly affects people of Northern European origin.

Haemochromatosis is caused by a mutation to the HFE gene. The HFE gene provides instructions for producing a protein which regulates iron levels in liver cells. The mutant allele is recessive to the normal allele.

(a) (i) State the meaning of the following terms:

I. gene; [1]

.....

.....

II. mutation. [1]

.....

.....

(ii) State the scientific technique which could be used to identify the presence of the mutant HFE allele. [1]

.....



(b) **Table 8.1** shows the amino acids coded for by each triplet code of DNA.

Table 8.1

		Second letter				Third letter
		T	C	A	G	
First letter	T	TTT } phe TTC } TTA } leu TTG }	TCT } TCC } ser TCA } TCG }	TAT } tyr TAC } TAA stop TAG stop	TGT } cys TGC } TGA stop TGG trp	
	C	CTT } CTC } leu CTA } CTG }	CCT } CCC } pro CCA } CCG }	CAT } his CAC } CAA } gln CAG }	CGT } CGC } arg CGA } CGG }	
	A	ATT } ATC } ile ATA } ATG met	ACT } ACC } thr ACA } ACG }	AAT } asn AAC } AAA } lys AAG }	AGT } ser AGC } AGA } arg AGG }	
	G	GTT } GTC } val GTA } GTG }	GCT } GCC } ala GCA } GCG }	GAT } asp GAC } GAA } glu GAG }	GCT } GGC } gly GGA } GGG }	

The DNA base sequences in **Image 8.2** show part of the normal and mutant HFE alleles found at the same position on a pair of chromosomes and part of the order of amino acids for which they code.

- (i) Use the information provided in **Table 8.1** to insert the missing amino acids in **Image 8.2**. [2]

Image 8.2

Normal allele	C T T	T G T	A G T
Amino acid sequence	leu	-	
Mutant allele	C T T	T A T	A G T
Amino acid sequence	leu	-	

- (ii) Explain how the mutant allele will give rise to haemochromatosis. [3]

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Examiner
only

- (c) (i) A woman, who is a carrier of the HFE mutation, and a man, who is affected by the condition are expecting a child.

Complete the Punnett square below to show the possible genotypes of their child. Use H to represent the normal allele.

[4]

Woman

Man

Phenotype ×

Genotype x

- (ii) State the probability of the couple having a child affected by haemochromatosis.

[1]

Probability =



Examiner
only

- (d) Genetic studies suggest the original mutant HFE allele arose in a single person who lived 60 to 70 generations ago. At that time, there was much less iron in people's diet.

Use the theory of natural selection to explain how the genetic condition haemochromatosis spread throughout the Northern European population at that time.

[4]

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TURN OVER FOR QUESTION 9



9. (a) In 2018–19, there was an outbreak of the Ebola virus disease in the Democratic Republic of the Congo (DRC). A double-blind clinical trial was carried out in four villages on 499 people infected with the Ebola virus. The purpose was to test the effectiveness of four potential treatments against the virus.

Table 9 shows the results halfway through the trial in August 2019.

Table 9

Treatment	Percentage of infected people dying from Ebola following the treatment (%)
REGN-EB3	29
mAb-114	33
ZMapp	49
Remdesivir	53

At the time of the trial, 1900 people out of 2831 confirmed cases of Ebola in other parts of the DRC had died of the disease. None of the 2831 people were part of the trial and so did not receive any of the four treatments above.

- (i) Use the information given above to calculate the percentage of confirmed cases of Ebola who received no treatment and died of the disease. **Give your answer to two significant figures.** [2]

Percentage =

- (ii) The results shown in **Table 9** convinced the scientists to stop using ZMapp and Remdesivir, and place all remaining patients on either REGN-EB3 or mAb-114.

Suggest why the scientists stopped using ZMapp and Remdesivir in August 2019 and placed all the remaining patients on either REGN-EB3 or mAb-114. [1]

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- (iii) Explain the meaning and importance of double-blind trials. [2]

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- (iv) This trial did not use a placebo. Suggest why a placebo could be considered unethical in this trial.

[1]

- (b) mAb-114 is a monoclonal antibody. It was developed using blood samples isolated from an Ebola survivor in 1995.

Outline the pre-clinical stages, in the correct sequence, which would have been carried out before the treatments could be used for clinical trials.

Describe how the monoclonal antibody mAb-114 could have been produced using lymphocytes from a sample of survivor's blood. [6]

[6 QER]

END OF PAPER

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